Finite Element method in Electromagnetics Applications using EMWorks software tools

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Contents

1 Overview **1**

1.1 Introduction to the finite element method in electromagnetic 1

1.1.1 What is the finite element method? 1

1.1.2 Introduction to Electromagnetism 11

1.1.3 Example:Electrostatic Problem 12

2 Examples Using EMWorks Software tools **22**

2.1 First Example :2D Electrostatics 22

2.2 Second Example :Magnetostatics problem 30

2.3 Third Example : Two movable magnets 37

2.4 fourth Example : Transient Magnetic simulation of a 3D Electric Motor . . 45

2.4.1 The band Component 47

2.4.2 Motion setup 47

2.4.3 Creating a Transient magnetic study 48

2.4.4 Winding configuration 50

2.4.5 Simulation 53

2.5 Fifth example :From 3D to 2D 58

2.5.1 2D simplification 59

2.5.2 Motion setup in EMworks2D 61

2.5.3 Band and air region 63

2.5.4 Transient study definition 64

2.6 RZ Problem with circuit 65

2.7 Conclusion 66

Chapter 1

Overview

1.1 Introduction to the finite element method in electromagnetic

1.1.1 What is the finite element method?

All we want is to solve a partial differential equation.

Let's say that we are given an equation such as

$$-\Delta u(x,y) = f(x,y) \text{ in } \Omega \subset \mathbb{R}^2$$
$$u(x,y) = 0 \text{ on } \partial\Omega$$

Where Ω is a surface and $\partial\Omega$ is its boundary.

It is the Laplace **P**artial **D**ifferential **E**quation(**PDE**).
The solution is a function, $u(x,y)$ and to know $u(x,y)$ means to know its value at infinitely many points (x,y).

But computers can only store finitely much data due to the memory constraint which means ,we can't exactly solve PDE's we can only approximate the exact solution.

Thus solving PDE's comes down to the following two questions :
1- What is the good way to approximate functions that requires only finitely much data?
2-How do we find an approximate solution to a PDE without knowing the exact solution itself?

There many ways to approximate functions(answer to question 1) :
1-Finite(Fourier series)

2-A global polynomial
3-Local(piecewise) polynomials defined on a subdivision (the mesh)of the domain Ω
4-...
For many good reasons, the finite element method uses option 3, i.e. piecewise polynomial approximation.

****Why not Fourier approximation ?**
Finite Fourier series of the form

$$u(x) \approx \sum_{k=1}^{N} U_k sin(kx) \tag{1.1}$$

are good approximations only if u(x) is globally smooth(Class C^1)
Example

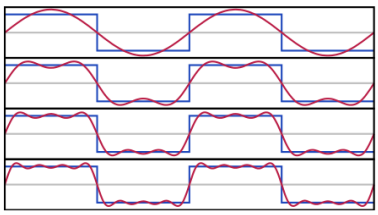


Figure 1.1: Fourier's approximation of a periodic square function

****Why not Taylor approximation ?**
Finite polynomial series of the form

$$u(x) \approx \sum_{k=0}^{N} U_k (x - x_0)^k \tag{1.2}$$

are good approximation only in neighbourhood
Example :

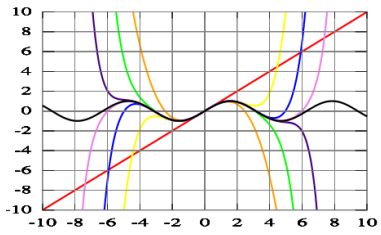


Figure 1.2: Fourier's approximation of a periodic square function

Solution : Piecewise approximation

- Split the domain on which you want to approximate u(x) into small parts(meshing) .
- Approximate separately on each part.

Small intervals $\longrightarrow$ Good approximation and insensitive to singularities
We can increase the resolution where necessary, This called h-adaptive mesh refinement it's a way to make computation cheaper

How do we find an approximation of the solution of a PDE without knowing the solution itself ?
Let's assume the following situation :

- We are in 1D
- We want to solve

$$-\frac{d^2}{dx^2}u(x) = f(x) \tag{1.3}$$

- We want a piecewise linear approximation of the solution u(x)
- We will call the approximation $u_h(x)$

$u_h(x)$ is a piecewise linear function and can be written as follows :

$$u_h(x) = \sum_{j=1}^{N} U_j \phi_j(x) \tag{1.4}$$

$\phi_j(x)$ are shape functions, they are known because we choose them, What remains is U_j ,in other words to know $u_h(x)$ we need to know the coefficients U_j

How do we find those coefficients ?
We need to use the PDE, equation 1.3

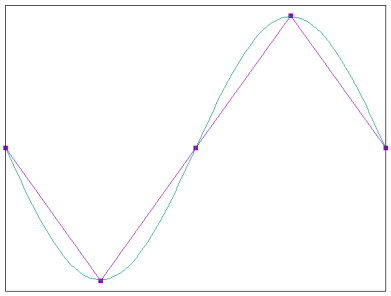
Idea 1:
Take the form :

$$u_h(x) = \sum_{j=1}^{N} U_j \phi_j(x) \tag{1.5}$$

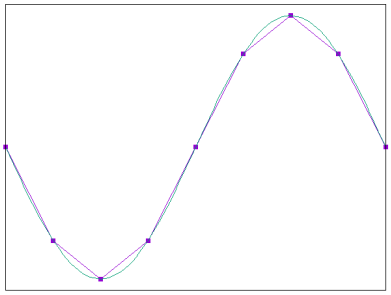
and put it into eq 1.3 :

$$-\frac{d^2}{dx^2}u_h(x) = -\frac{d^2}{dx^2}\sum_{j=1}^{N} U_j \phi_j(x) = f(x) \tag{1.6}$$

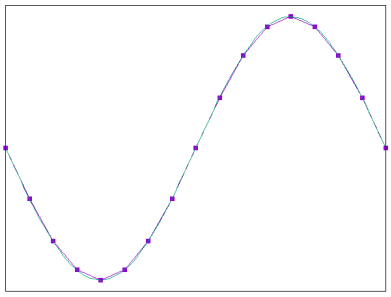
This does not work !



(a) poor approximation

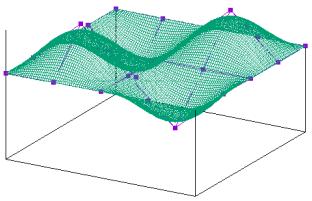


(b) better approximation

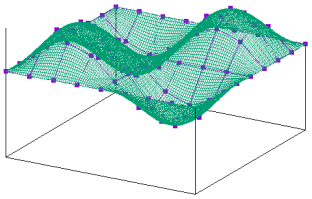


(c) excellent approximatin

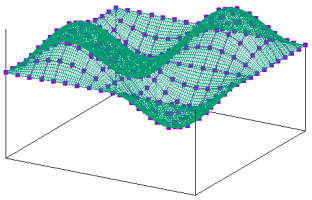
Figure 1.3: Three graphs showing the approximation of a 1D function (maybe the solution to a single variable PDE)



(a) poor approximation

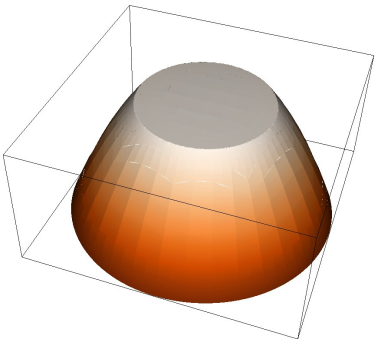


(b) better approximation

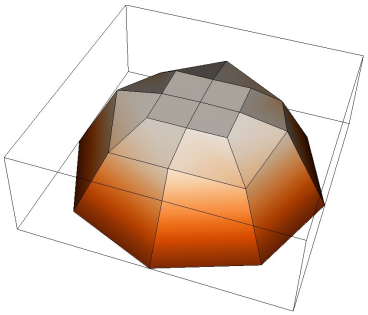


(c) excellent approximatin

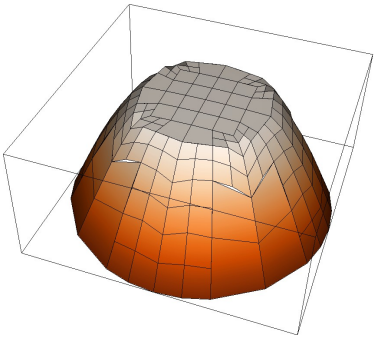
Figure 1.4: Three graphs showing the approximation of a 2D function (maybe the solution to two variables PDE)



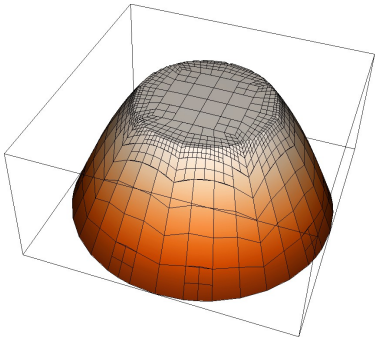
(a) Exact Solution



(b) approximation on a coarse mesh



(c) approximation on a somehow refined mesh



(d) approximation on a decent mesh

Figure 1.5: Exact solution and approximated solution in different mesh grids

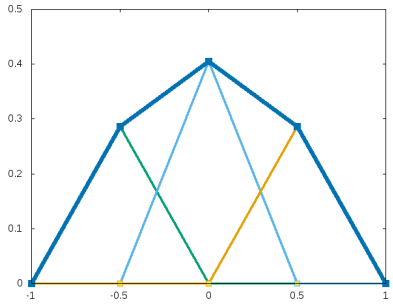


Figure 1.6: Approx

- Second derivatives are zero on each interval since $\phi_j(x)$ are first order linear functions
- Second derivatives are not defined at node points

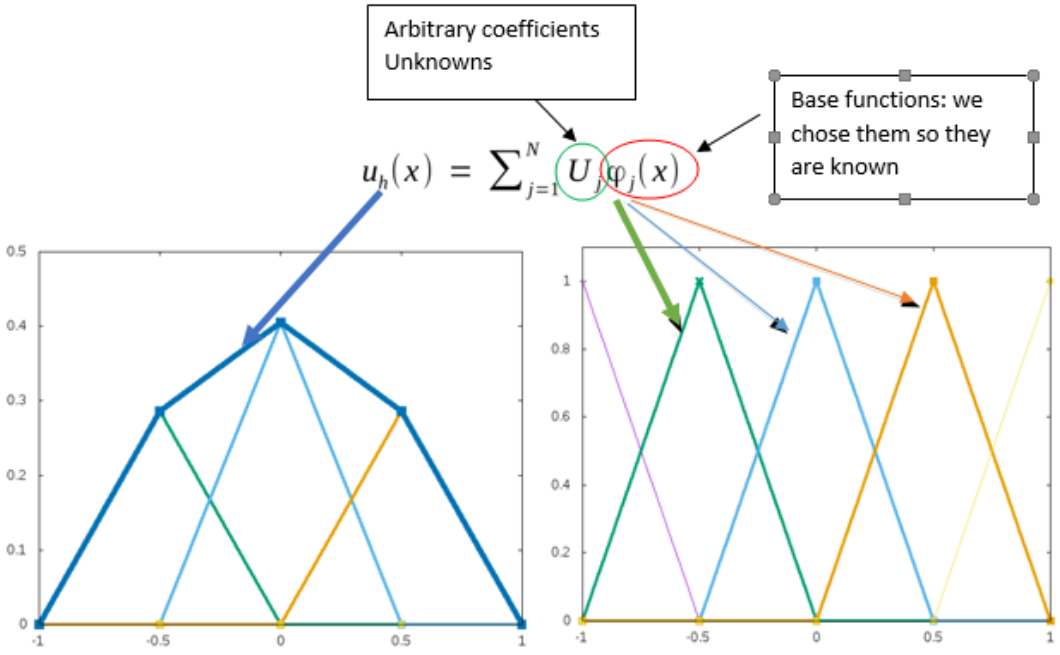


Figure 1.7: Piecewise linear approx

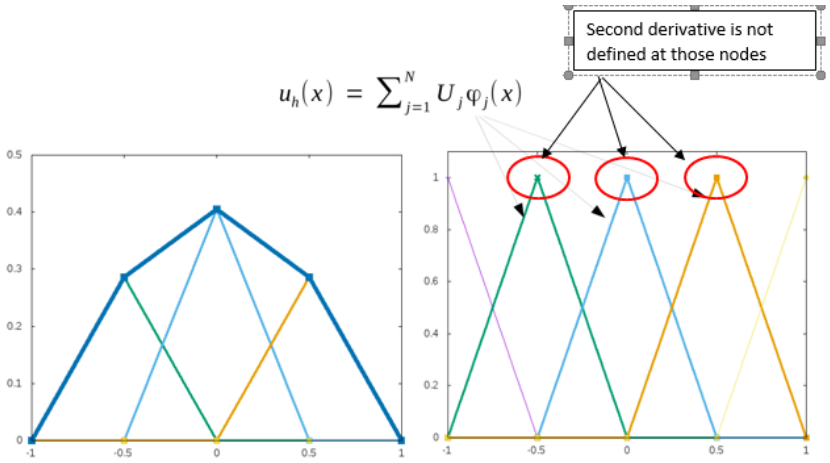


Figure 1.8: shape function nodes

Idea 2 : Use the Mathematical theory of weak solutions

Starting point of this theory When we say that two functions g(x) and h(x) need to be equal,

$$g = h \tag{1.7}$$

- We mean that g and h are equal for every x?

- We mean that g and h are equal for almost every x ?
- . . . ?

This problem arises because we only know how to compare numbers, but now we need to compare functions !

A solution

Turn the equation

$$g = h \tag{1.8}$$

into an infinite number of comparisons of numbers :

$$\begin{aligned} F_1[g] &= F_1[h] \\ F_2[g] &= F_2[h] \\ F_3[g] &= F_3[h] \\ &\vdots \end{aligned}$$

for a infinite number of appropriate functionals $F_i[.]$

Example :

We say that g equals h in the interval [0,1] if

$$\begin{aligned} \int_0^1 g(x)dx &= \int_0^1 h(x)dx \\ \int_0^1 xg(x)dx &= \int_0^1 xh(x)dx \\ \int_0^1 x^2g(x)dx &= \int_0^1 x^2h(x)dx \\ &\dots \end{aligned}$$

For differential equations

For differential equations, we say $u(x)$ is a weak solution of a PDE if

$$\begin{aligned} F_1[\frac{-d^2}{dx^2}u(x)] &= F_1[f(x)] \\ F_2[\frac{-d^2}{dx^2}u(x)] &= F_2[f(x)] \\ F_3[\frac{-d^2}{dx^2}u(x)] &= F_3[f(x)] \\ &\dots \end{aligned}$$

for infinitely functional F_i
Here we can choose F_i to be

$$F_k(g(x)) = \int_\Omega \phi_k(x) g(x) d\Omega \tag{1.9}$$

i.e. in our case :

$$\int_\Omega \phi_k(x)[-\frac{d^2}{dx^2}u(x)]dx = \int_\Omega \phi_k(x) f(x) dx \tag{1.10}$$

This equation holds for all test functions $\phi(x)$

We can integrate by parts to get :

$$\int_\Omega \phi(x)[-\frac{d^2}{dx^2}u(x)]dx = \int_\Omega [\frac{d}{dx}\phi(x)][\frac{d}{dx}u(x)]dx + \text{Boundary Terms} \tag{1.11}$$

Now the exact solution $u(x)$ satisfies 1.11 we can try to find an approximate solution $u_h(x)$ that satisfies 1.11 in that case we have

$$\int_\Omega \phi_k(x)[-\frac{d^2}{dx^2}u_h(x)]dx = \int_\Omega [\frac{d}{dx}\phi_k(x)][\frac{d}{dx}u_h(x)]dx + \text{Boundary Terms} \tag{1.12}$$

So we seek a solution $u_h(x) = \sum_{j=1}^{N} U_j \phi_j(x)$ that satisfies equation 1.12 for an infinite test functions $\phi_k(x)$.
with that being said, we will have an N number of unknowns U_j and an infinite number of equations.

Idea 3

Since the number of test functions $\phi_k(x)$ is infinite, we need to restrict its number to get an approximate solution
i.e.
We seek

$$u_h(x) = \sum_{j=1}^{N} U_j \phi_j(x) \tag{1.13}$$

So that

$$\int_\Omega [\frac{d}{dx}\phi_h(x)][\frac{d}{dx}u_h(x)]dx = \int_\Omega \phi_h(x) f(x) dx \tag{1.14}$$

$\forall \phi_h = \sum_{k=1}^{N} c_k \phi_k(x)$
This equivalent to N equations for N unknowns and it's called the Galerkin Method

in this method, the test functions are the same as the base functions used in u_h

we can use the scalar product notation, so we get

$$(g,h)_\Omega = \int_\Omega g(x)h(x)dx \tag{1.15}$$

and the problem can be rewritten

$$\int_\Omega [\frac{d}{dx}\phi_h(x)][\frac{d}{dx}u_h(x)]dx = (\frac{d}{dx}\phi_h(x), \frac{d}{dx}u_h(x))_\Omega = (\phi_h(x), f(x))_\Omega \tag{1.16}$$

$\forall \phi_h = \sum_{k=1}^{N} c_k \phi_k(x)$

i.e. we seek an approximate solution $u_h(x) = \sum_{j=1}^{N} U_j \phi_j(x)$ so that :

$$\begin{aligned}
(\frac{d}{dx}\phi_1(x), \frac{d}{dx}u_h(x))_\Omega &= (\phi_1(x), f(x))_\Omega \\
(\frac{d}{dx}\phi_2(x), \frac{d}{dx}u_h(x))_\Omega &= (\phi_2(x), f(x))_\Omega \\
(\frac{d}{dx}\phi_3(x), \frac{d}{dx}u_h(x))_\Omega &= (\phi_3(x), f(x))_\Omega \\
&\dots
\end{aligned}$$

Similarly, in higher dimensions, the problem will be as follows :
start with (also for Laplace PDE)

$$-\Delta u(\vec{x}) = f(\vec{x}) \tag{1.17}$$

Multiply by a test function and integrate

$$\int_\Omega \phi(\vec{x})[-\Delta u(\vec{x})]d\Omega = \int_\Omega \phi(\vec{x}) f(\vec{x}) \tag{1.18}$$

Now integrate by parts the left-hand side to eliminate the need for second derivatives (check Green's second identity)

$$\int_\Omega \phi(\vec{x})[-\Delta u(\vec{x})]d\Omega = \int_\Omega \nabla\phi(\vec{x})\nabla u(\vec{x}) + \text{Boundary Terms} \tag{1.19}$$

We then seek :

$$u_h(x) = \sum_{j=1}^{N} U_j \phi_j(\vec{x}) \tag{1.20}$$

so that

$$(\nabla\phi_h, \nabla u_h) = (\phi_h, f)\ \forall\phi_h = \sum_{k=1}^{N} c_k\phi_k \tag{1.21}$$

Source (Wolfgang Bangerth course)

1.1.2 Introduction to Electromagnetism

Fundamental equations summary

The electromagnetic theory is governed by Maxwell's equations :

$$\boldsymbol{\nabla} \times \vec{H}(P,t) = \vec{J} + \frac{\partial \vec{D}(P,t)}{\partial t} \tag{1.22}$$

$$\boldsymbol{\nabla} \times \vec{E}(P,t) = -\frac{\partial \vec{B}(P,t)}{\partial t} \tag{1.23}$$

$$\boldsymbol{\nabla} \cdot \vec{B}(P,t) = 0 \tag{1.24}$$

$$\boldsymbol{\nabla} \cdot \vec{D}(P,t) = \rho(P,t) \tag{1.25}$$

where :
$\vec{H}(P,t)$ is the magnetic field strength
$\vec{D}(P,t)$ is the electric displacement
$\vec{E}(P,t)$ is the electric field strength
$\vec{B}(P,t)$ is the magnetic flux density
There are also the constitutive relationships and the continuity equation :

$$\vec{B}(P,t) = \mu(P)\vec{H}(P,t) \tag{1.26}$$

$$\vec{D}(P,t) = \epsilon(P)\vec{E}(P,t) \tag{1.27}$$

$$\vec{J}(P,t) = \sigma(P)\vec{E}(P,t) \tag{1.28}$$

and the continuity equation :

$$\boldsymbol{\nabla} \cdot \vec{J}(P,t) = -\frac{\partial \rho(P,t)}{\partial t} \tag{1.29}$$

Flux density is written in terms of the vector potential, A, as:

$$\vec{B}(P,t) = \boldsymbol{\nabla} \times A(P,t) \tag{1.30}$$

$A(P,t)$ is the magnetic vector potential

For low frequency problems, displacement current can be ignored

1.1.3 Example:Electrostatic Problem

Consider a parallel plate capacitor with a uniform dielectric of permittivity ϵ between the plates

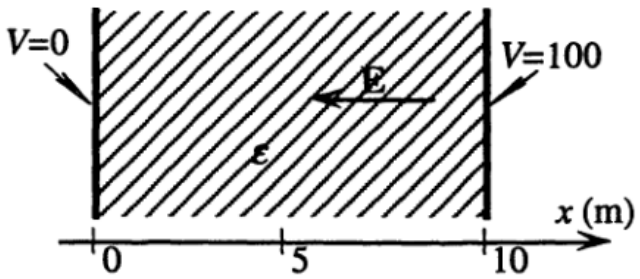


Figure 1.9: Parallel plate capacitor

Assuming the electric field has only an x component, the problem is one dimensional.

The potential between the plates varies linearly from V=0 at x=0 and V=100V at x=10m we will try to find the Electric field E

Analytical solution

Since we assumed that the problem is electrostatic, there is no time dependence for the electric field. equations 1.23 and 1.25 can be used to derive the Electric filed expression

$$\boldsymbol{\nabla} \times E = 0 \tag{1.31}$$

this equation gives

$$E = -\nabla V \tag{1.32}$$

$$\boldsymbol{\nabla} \cdot D = \rho \tag{1.33}$$

and using the constitutive relation ,equation 1.27

$$D = \epsilon E \tag{1.34}$$

replacing in 1.33 gives

$$\boldsymbol{\nabla} \cdot \epsilon E = \boldsymbol{\nabla} \cdot \epsilon \nabla V = -\rho \tag{1.35}$$

which in explicit form is

$$\frac{\partial}{\partial x}\epsilon\frac{\partial V}{\partial x} + \frac{\partial}{\partial y}\epsilon\frac{\partial V}{\partial y} + \frac{\partial}{\partial z}\epsilon\frac{\partial V}{\partial z} = -\rho \tag{1.36}$$

If there are no static charges ($\rho = 0$) and a single dielectric material exists in the domain, and we are in 1D , the equation becomes

$$\frac{\partial^2 V}{\partial^2 x} = 0 \tag{1.37}$$

by direct integration we can get the solution

$$V(x) = ax + b \tag{1.38}$$

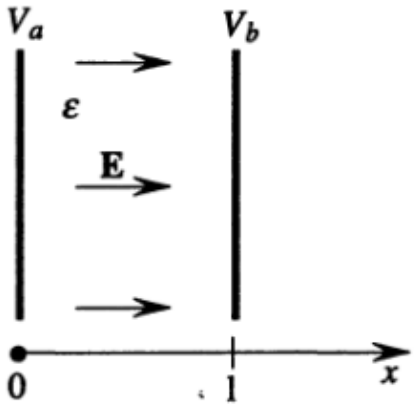


Figure 1.10: Electric field

and using the boundary conditions, and $E = -\nabla V$ we get

$$E_x = \frac{V_a - V_b}{l} \tag{1.39}$$

in our case we have

$$\vec{E} = -10\vec{i} \tag{1.40}$$

The finite element solution

a. Step 1 - The establishment of the physical equations

The equation that we want to solve is the Laplace equation

$$\boldsymbol{\nabla} \cdot \epsilon \nabla V = -\rho \tag{1.41}$$

in 1D it becomes

$$(\frac{\partial}{\partial x}\vec{i}).\epsilon(\frac{\partial V}{\partial x}\vec{i}) = -\rho \tag{1.42}$$

and the set of boundary conditions $V(0) = 0$ and $V(10) = 100V$

The exact solution is known from the previous computation, i.e.

$$\vec{E} = -10\vec{i}[V/m] \tag{1.43}$$

b. Step 2 - Discretization and finite element modelling

In 1D applications, the line representing the solution domain is cut into small segments, and each segment is called a "finite element." The points defining the element are called "nodes" or "degrees of freedom.

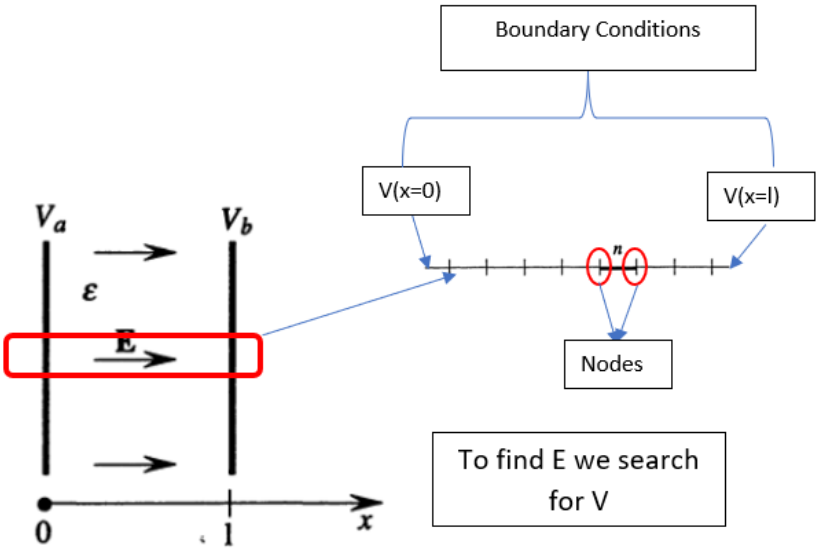


Figure 1.11: mesh nodes in 2D domain

in 1D it looks like this

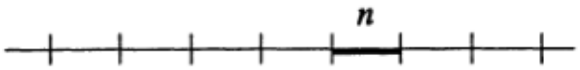


Figure 1.12: 1D mesh

in our PDE

$$\frac{\partial}{\partial x}\epsilon\frac{\partial V}{\partial x} = -\rho \tag{1.44}$$

We want to search for an approximate solution to V like previously we will use piece wise linear approximation i.e. We seek V_h

$$V_h(x) = \sum_{j=1}^{N} V_j\phi_j(x) \tag{1.45}$$

We can assume that inside an element the potential varies linearly

$$V(x) = a_1 + a_2x \tag{1.46}$$

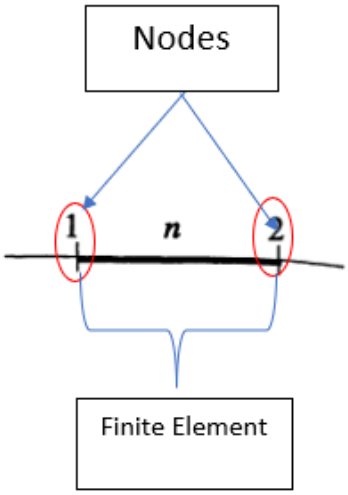


Figure 1.13: 1D finite element and its nodes

this gives at node number 1 $x = x_1$

$$V(x_1) = V_1 = a_1 + a_2 x_1 \tag{1.47}$$

and at node 2 $x = x_2$

$$V(x_2) = V_2 = a_1 + a_2 x_2 \tag{1.48}$$

We can find $a_1 \text{and} a_2$ by solving the linear system

$$\begin{aligned} V_1 &= a_1 + a_2 x_1 \\ V_2 &= a_1 + a_2 x_2 \end{aligned}$$

we get

$$\begin{aligned} a_1 &= \frac{V_1 x_2 - V_2 x_1}{x_2 - x_1} \\ a_2 &= \frac{V_2 - V_1}{x_2 - x_1} \end{aligned}$$

denoting the length of the element as $h = x_2 - x_1$

$$\begin{aligned} a_1 &= \frac{1}{h}(V_1 x_2 - V_2 x_1) \\ a_2 &= \frac{1}{h}(V_2 - V_1) \end{aligned}$$

Substituting in equation 1.45

$$V(x) = \frac{1}{h}[V_1 x_2 - V_2 x_1 + (V_2 - V_1)x] = \frac{1}{h}[(x_2 - x)V_1 + (-x_1 + x)V_2] \tag{1.49}$$

or

$$V(x) = \sum_{i=1,2} \frac{1}{h}(p_i + q_i x)V_i \tag{1.50}$$

where : $p_1 = x_2$ $q_1 = -1$ $p_2 = -x_1$ and $q_2 = 1$ which can be also written as in equation 1.44

$$V(x) = \sum_{i=1,2} \phi_i(x)V_i \tag{1.51}$$

We search for V_i

c.Step 3 - Application of the weak formulation

Now it is time to distinguish between the "exact solution," V, and the solution obtained with the finite element method V_h

using our approximated solution inside our PDE generates a Residual

$$\frac{\partial}{\partial x}\epsilon\frac{\partial V_h}{\partial x} + \rho = R \tag{1.52}$$

we can use a test function ϕ (also called weighting function) such that

$$\int_\Omega \phi(x)[\frac{\partial}{\partial x}\epsilon\frac{\partial V_h}{\partial x} + \rho]d\Omega = 0 \tag{1.53}$$

$$\int_\Omega \phi(x)[\frac{\partial}{\partial x}\epsilon\frac{\partial V_h}{\partial x} + \rho]d\Omega = \int_\Omega \phi(x)[\frac{\partial}{\partial x}\epsilon\frac{\partial V_h}{\partial x}]d\Omega + \int_\Omega \phi(x)\rho d\Omega \tag{1.54}$$

and now we integrate by parts to get rid of second derivatives(so we get the weak form of our PDE) using

$$\int UV' = [UV]_{Boundaries} - \int U'V \tag{1.55}$$

with $U = \phi$ and $V = \epsilon\frac{\partial V_h}{\partial x}$

$$\int_\Omega \phi(x)[\frac{\partial}{\partial x}\epsilon\frac{\partial V_h}{\partial x}]d\Omega = [\phi(x)\epsilon\frac{\partial V_h(x)}{\partial x}]_{Boundaries} - \int_\Omega \epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial \phi(x)}{\partial x}d\Omega \tag{1.56}$$

in that equation we have $[\phi(x)\epsilon\frac{\partial V_h(x)}{\partial x}]_{Boundaries} = 0$ so that .

$$\int_\Omega \phi(x)[\frac{\partial}{\partial x}\epsilon\frac{\partial V_h(x)}{\partial x}]d\Omega = -\int_\Omega \epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial \phi(x)}{\partial x}d\Omega \tag{1.57}$$

so equation 1.52 becomes

$$-\int_{\Omega}\epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial \phi(x)}{\partial x}d\Omega+\int_{\Omega}\phi(x)\rho d\Omega=0 \tag{1.58}$$

or

$$\int_{\Omega}[\epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial \phi(x)}{\partial x}-\phi(x)\rho]d\Omega=0 \tag{1.59}$$

in our mesh (discrete domain) this equation becomes

$$\sum_{n=1,N}\int_{\Omega_n}[\epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial \phi(x)}{\partial x}-\phi(x)\rho]d\Omega_n=0 \tag{1.60}$$

in our mesh Ω_n is the element and its dimension is $h=x_2-x_1$,N is the number of elements
We have from equation 1.49

$$V_h(x)=\frac{1}{h}(p_1+q_1x)V_1+\frac{1}{h}(p_2+q_2x)V_2 \tag{1.61}$$

from this we have

$$\frac{\partial V_h}{\partial x}=\frac{1}{h}q_1V_1+\frac{1}{h}q_2V_2 \tag{1.62}$$

also for ϕ which is a combination of ϕ_1 and ϕ_2

$$\frac{\partial \phi_1}{\partial x}=\frac{\partial}{\partial x}(\frac{1}{h}(p_1+q_1x))=\frac{1}{h}q_1 \tag{1.63}$$

same for ϕ_2

$$\frac{\partial \phi_2}{\partial x}=\frac{1}{h}q_2 \tag{1.64}$$

from equation 1.41 we had a scalar product
so substituting in equation 1.60 we have for first left term

$$\int_{\Omega_n}[\epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial \phi(x)}{\partial x}-\phi(x)\rho]d\Omega_n \tag{1.65}$$

$$\int_{\Omega_n}[\epsilon[\frac{\partial V_h(x)}{\partial x}\vec{i}].[\frac{\partial \phi(x)}{\partial x}\vec{i}]d\Omega_n \tag{1.66}$$

which is a system of 2 equations for node 1 and node 2

$$\int_h\epsilon([\frac{1}{h}q_1V_1+\frac{1}{h}q_2V_2]\vec{i}).([\frac{1}{h}q_1]\vec{i})dx$$
$$\int_h\epsilon([\frac{1}{h}q_1V_1+\frac{1}{h}q_2V_2]\vec{i}).([\frac{1}{h}q_2]\vec{i})dx$$

i.e.

$$\int_h \epsilon[\frac{1}{h^2}(q_1^2 V_1 + q_1 q_2 V_2])dx$$
$$\int_h \epsilon[\frac{1}{h^2}(q_1 q_2 V_1 + q_2^2 V_2])dx$$

since the integrand doesn't depend on x we get

$$[\frac{\epsilon}{h^2}(q_1^2 V_1 + q_1 q_2 V_2])(\int_h dx = h)$$
$$[\frac{\epsilon}{h^2}(q_1 q_2 V_1 + q_2^2 V_2])dx \int_h dx$$

which is

$$[\frac{\epsilon}{h}(q_1^2 V_1 + q_1 q_2 V_2])$$
$$[\frac{\epsilon}{h}(q_1 q_2 V_1 + q_2^2 V_2])dx$$

we can put it in a more compact form as a matrix

$$\frac{\epsilon}{h}\begin{bmatrix} q_1^2 & q_1 q_2 \\ q_1 q_2 & q_2^2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{1.67}$$

Now we evaluate the second term, and we consider ρ as constant in element number n

$$-\int_{\Omega_n} \phi(x)\rho d_{\Omega_n} = -\rho \int_h \phi(x)dx \tag{1.68}$$

inside an element this term $\int_h \phi(x)dx$ represents the area of a triangle with base h and height 1 which is equal to $\frac{h}{2}$ as represented in the below figure

in matrix form it becomes knowing that in each element the base function

$$\frac{-\rho h}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{1.69}$$

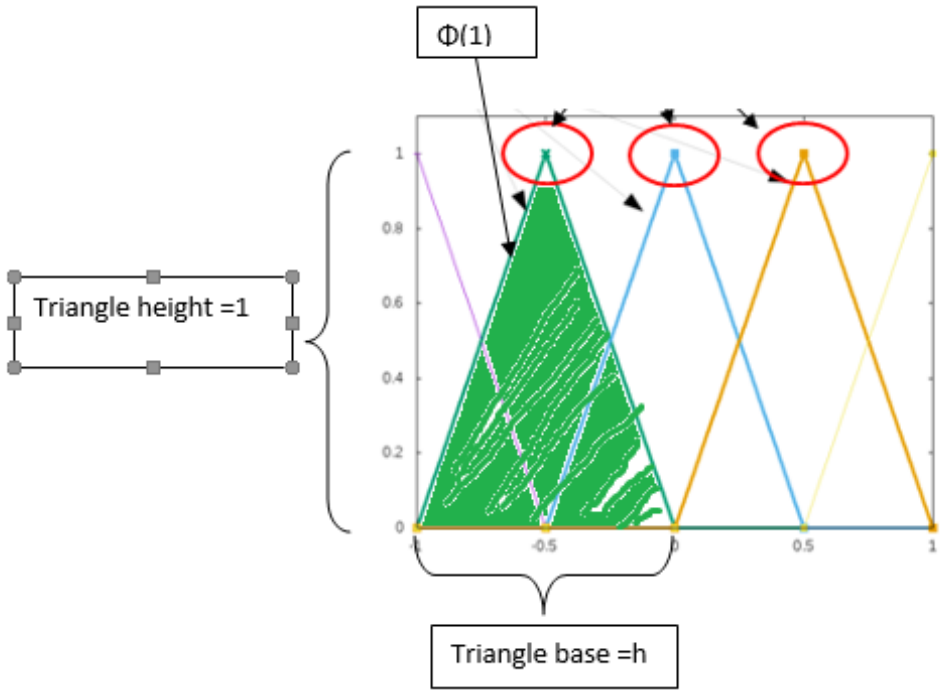


Figure 1.14: area of base function

**Now let's suppose that the domain is divided into 2 finite elements and that $\rho = 0$

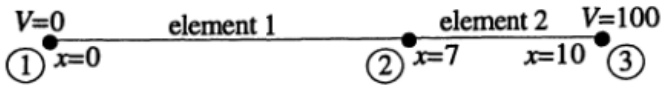


Figure 1.15: our domain discretized into two elements

at x=7 we don't know the potential V_2 so we have one unknown, at x =0 and x= 10 we know the potential (our boundary conditions)

for element number 1 we calculate our matrix coefficient in equation 1.67 knowing from page 16 equation 1.50 our coefficients

$p_1 = x_2$,$q_1 = -1$, $p_2 = -x_1$, and $q_2 = 1$

$$\frac{\epsilon}{h}\begin{bmatrix} q_1^2 & q_1 q_2 \\ q_1 q_2 & q_2^2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{1.70}$$

for element 1 $h = x_2 - x_1 = 7$ equation 1.70 becomes

$$\frac{\epsilon}{7}\begin{bmatrix} (-1)^2 & (-1)(1) \\ (-1)(1) & 1^2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \frac{\epsilon}{7}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{1.71}$$

Because there are three nodes in the solution domain, the matrix above, also called an "elemental matrix," must be placed or assembled into a larger "global matrix" which includes all the nodes in the system.
Doing so results in the following global matrix and equation 1.71 becomes for elelemt 1

$$\frac{\epsilon}{7}\begin{bmatrix}1 & -1 & 0\\ -1 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} \tag{1.72}$$

Evaluating now element number 2 we get the element matrix, $h = 3$

$$\frac{\epsilon}{3}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\begin{bmatrix}V_2\\ V_3\end{bmatrix} \tag{1.73}$$

also for element 2 we form the global matrix

$$\frac{\epsilon}{3}\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & -1\\ 0 & -1 & 1\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} \tag{1.74}$$

going back to equation 1.60

$$\sum_{n=1,3}\int_{\Omega_n}[\epsilon\frac{\partial V_h(x)}{\partial x}\frac{\partial\phi(x)}{\partial x} - \phi(x)\rho]d\Omega_n = 0 \tag{1.75}$$

we sum over all elements in process called assembly

$$\frac{\epsilon}{7}\begin{bmatrix}1 & -1 & 0\\ -1 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} + \frac{\epsilon}{3}\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & -1\\ 0 & -1 & 1\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix} \tag{1.76}$$

we get

$$\epsilon\begin{bmatrix}\frac{1}{7} & \frac{-1}{7} & 0\\ \frac{-1}{7} & \frac{1}{7}+\frac{1}{3} & \frac{-1}{3}\\ 0 & \frac{-1}{3} & \frac{1}{3}\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix} \tag{1.77}$$

Because the values of V_1 and V_3 are equal respectively to 0 and 100 we can reform the system so it will look like this $\epsilon\#0$

$$\begin{bmatrix}1 & 0 & 0\\ \frac{-1}{7} & \frac{10}{21} & \frac{-1}{3}\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 100\end{bmatrix} \tag{1.78}$$

this is the same as writing the second equation alone because the first and third equation gives the boundary conditions

$$\begin{cases} V_1 = 0 \\ -\frac{1}{7}V_1 + \frac{10}{21}V_2 + \frac{-1}{3}V_3 = 0 \\ V3 = 100 \end{cases} \tag{1.79}$$

with $V_1 = 0$ and $V_3 = 100$

$$V_2 = 70V \tag{1.80}$$

and now we can obtain the value of the electric field intensity, sum over elements

$$E = -\nabla V(x) = -\frac{\partial}{\partial x}\sum_{i=1,2}\frac{1}{h}(p_i + q_i x)V_i\vec{i} = -\frac{1}{h}(q_1V_1 + q_2V_2)\vec{i} \tag{1.81}$$

for element number 1 this gives

$$E = -\frac{1}{7}(-1 * 0 + 1 * 70)\vec{i} = -10\vec{i} \tag{1.82}$$

for element number 2 this gives

$$E = -\frac{1}{3}(-1 * 70 + 1 * 100)\vec{i} = -10\vec{i} \tag{1.83}$$

which is our analytical result page 13

and with this basic idea about the method, in 2D and 3D things are more complex to perform manually, but the same principles are used and with the advancement in technology we can appreciate software tools, we will be instead using EMWorks software tools to simulate 2D and 3D problems .

Chapter 2

Examples Using EMWorks Software tools

2.1 First Example :2D Electrostatics

In this example, we will use EMworks2D to simulate an electrostatics XY planar problem, so first we prepare the geometric model of our problem inside a CAD tool or maybe import the geometry, and then apply materials and boundary conditions as we've seen in the previous chapter, they are needed so the PDE can be solved, and we get a unique solution

in EMWorks2D, we have Two types of problems that can be treated,
1- XY planar symmetric problems
2- RZ axisymmetric problems

for XY planar symmetry, there should be an invariance in geometric and electric proprieties along the XY plane,
for RZ axisymmetric problems, there should be a rotational invariance along the Z axis for both geometric and electric proprieties

our first example is a planar capacitor as in the figure below :

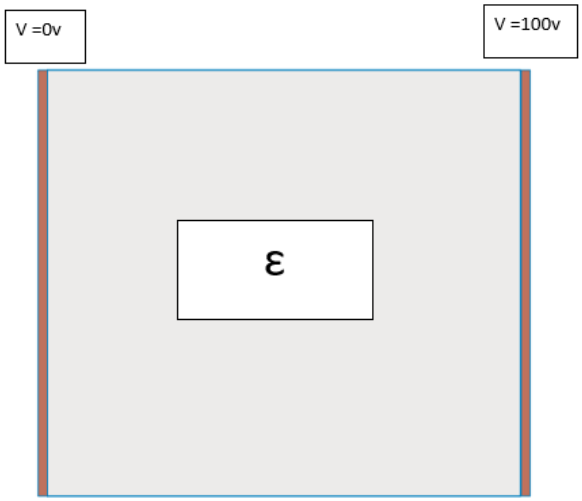


Figure 2.1: 2D planar capacitor

if you have SolidWorks opened, first thing we need to create a local coordinate system which is straightforward

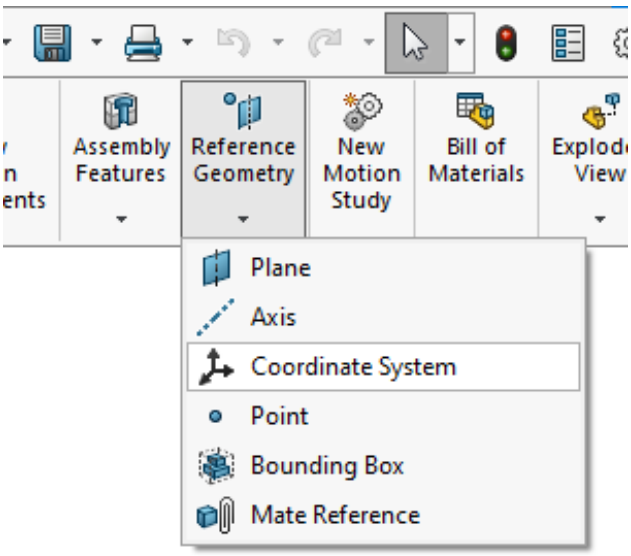


Figure 2.2: local coordinate systems

our model will be in the XY plane
2-next step we create an electrostatic study

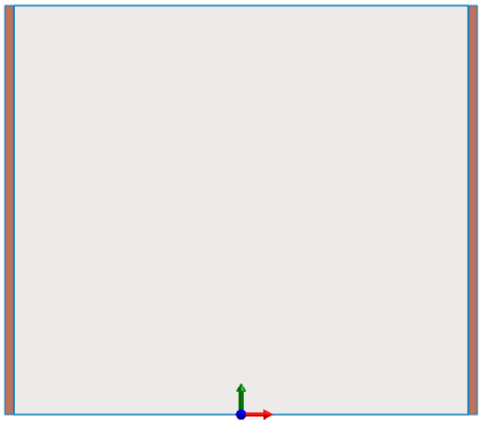


Figure 2.3: local coordinate systems

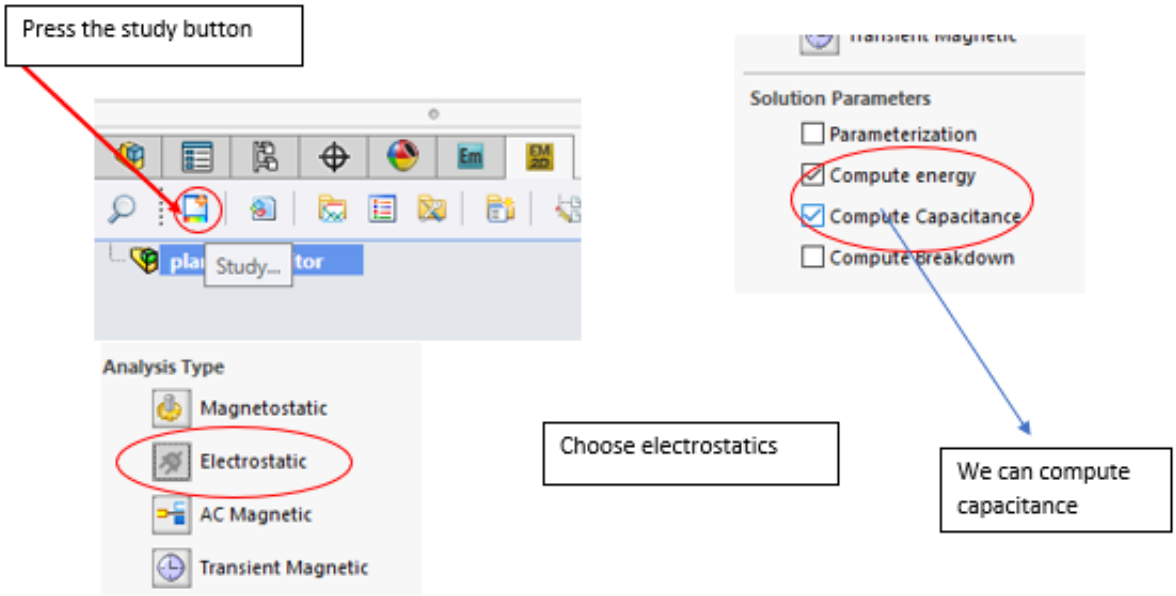


Figure 2.4: study definition

in the 2D geometry tab choose the type of problem either it is a planar symmetrical problem or RZ axisymmetric problem, choose the coordinate system and the model length in the case of planar symmetry problem, in our case it's an XY planar problem and the model length is set to be 1mm

after validation, we chose materials and boundary conditions

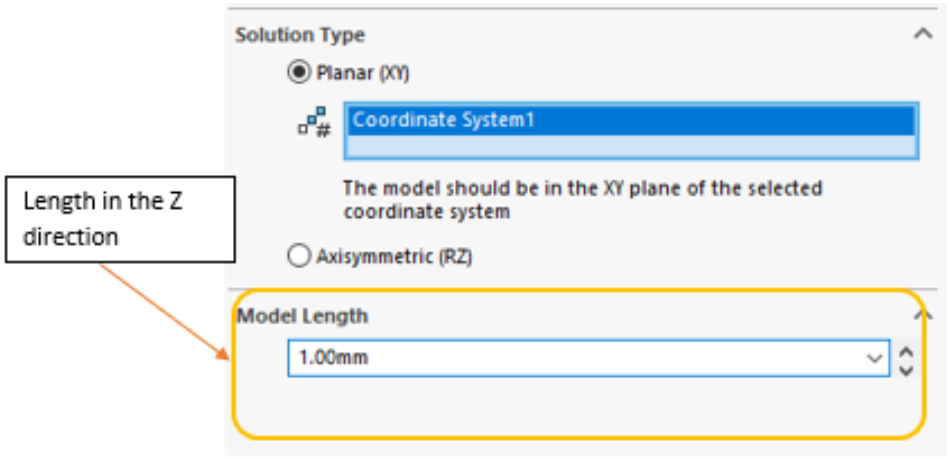


Figure 2.5: problem type

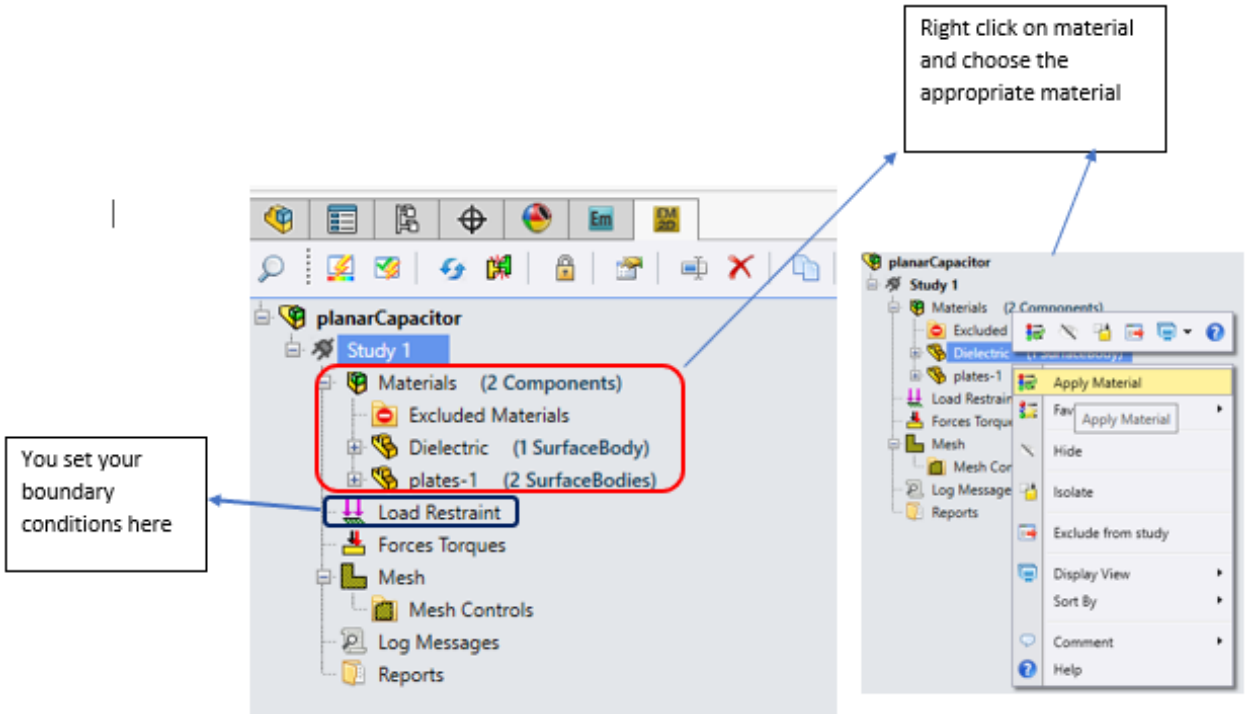


Figure 2.6: Materials and boundary conditions

in our planar capacitor, boundary conditions are fixed voltages in the two plates, so right click on load restraint and choose fixed voltage and set the two plates voltages to 0 and 100 volts

after setting up the study, we can validate before run to check if there is something missing and then run

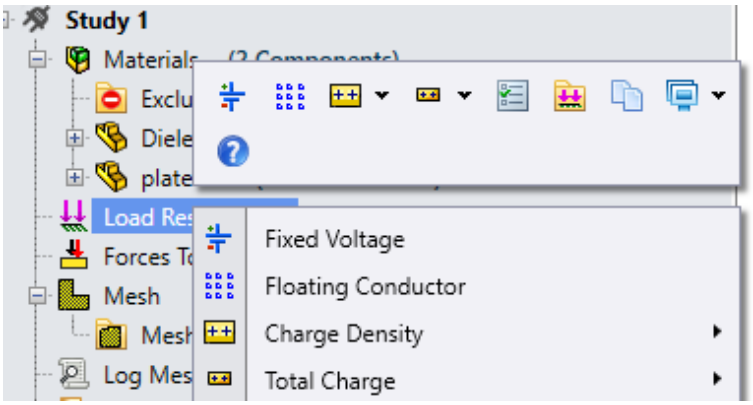


Figure 2.7: fixed voltage boundary condition

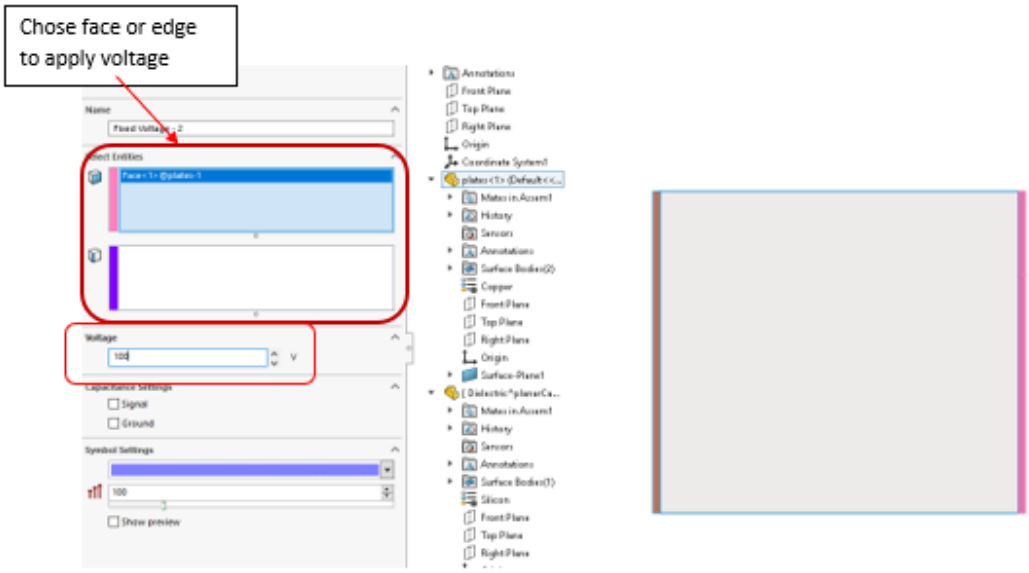


Figure 2.8: fixed voltage boundary condition

we can check the discretized domain to see where our PDE will be solved approximately

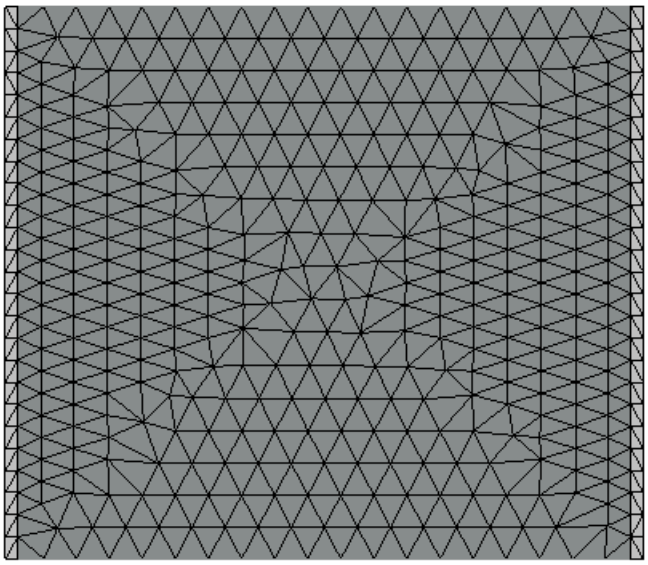


Figure 2.9: our domain discretized (mesh grid)

**Simulation results As w've seen before the electric field will be along the X direction, the field strength will be different as in our example before because the distance between the two plates is set to 100 mm instead of 10 m,the electric field is does depend on the inverse of a distance so its strength will be multiplied by 100, we guess to have

$$E = -1000\vec{i}\frac{V}{m} \tag{2.1}$$

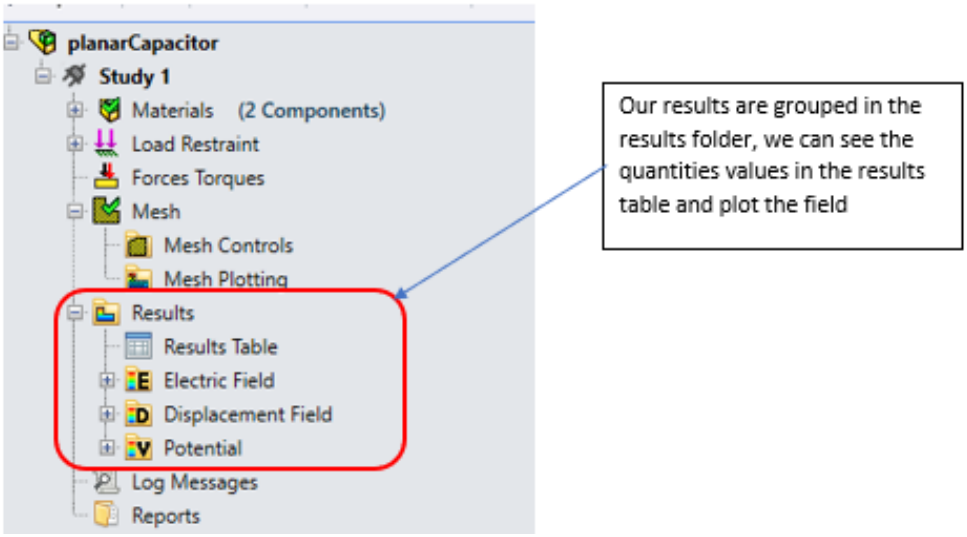


Figure 2.10: Results

we can plot the electric field by opening the Electric field folder and plotting or by right-clicking on the electric field and choosing vector plot as shown in the below figure.

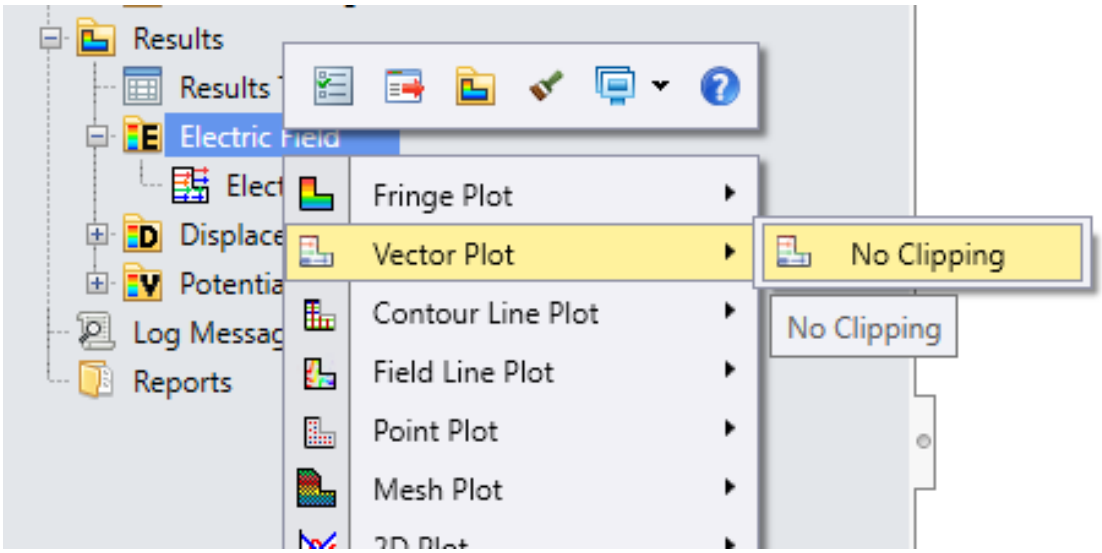


Figure 2.11: plotting the electric field

we get this result

we can also plot the change in the electric field intensity between any given tow points by using the 2D plot option

if we validate, we can see that the filed is constant and that what we expected it to be.

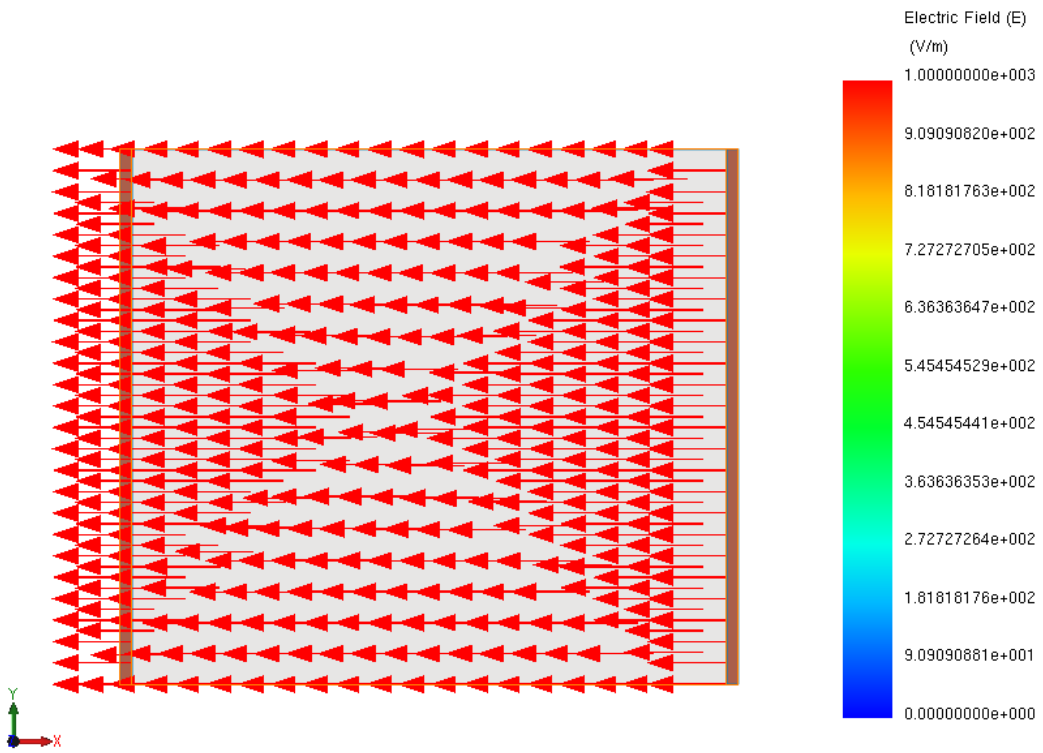


Figure 2.12: Electric field vector plot

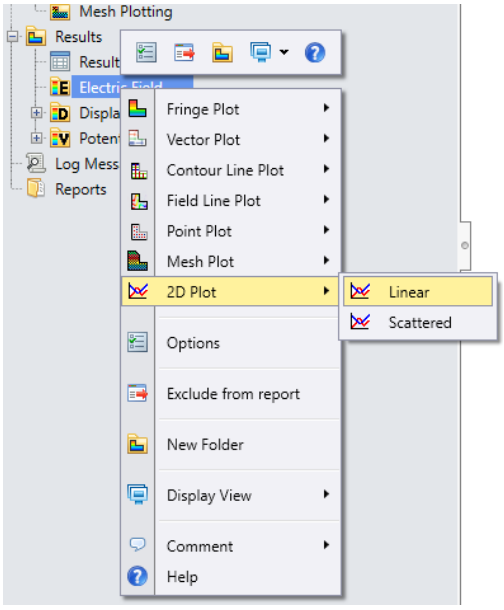


Figure 2.13: Electric field vector plot

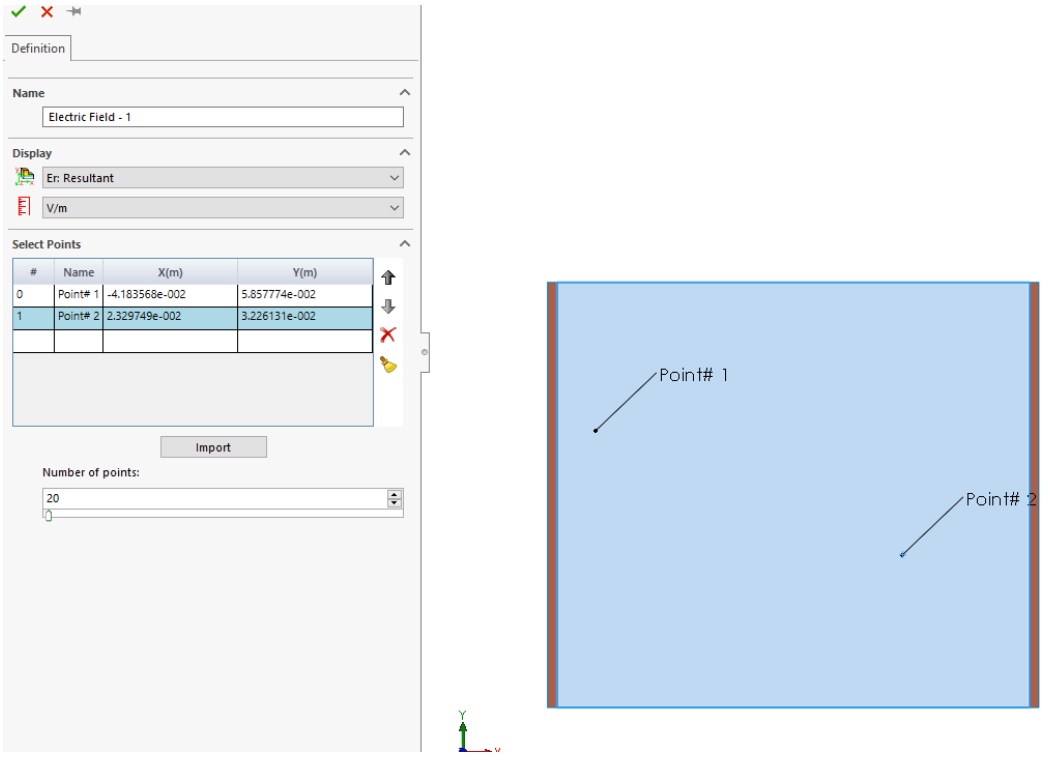


Figure 2.14: Electric field vector plot

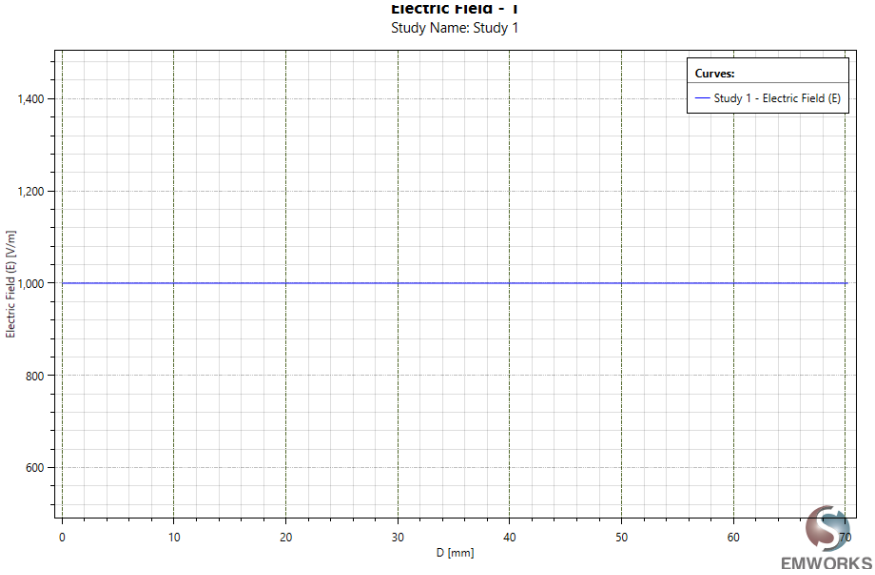


Figure 2.15: Electric field between two points

2.2 Second Example :Magnetostatics problem

As we did before, first thing we need to create our local coordinates system

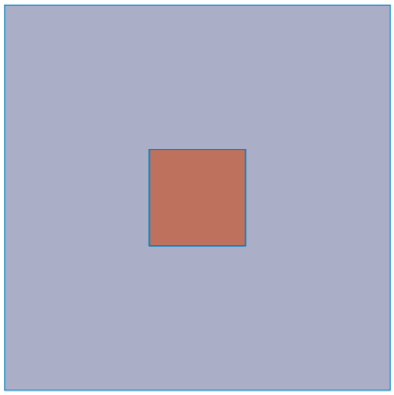


Figure 2.16: Current conductor 2D model

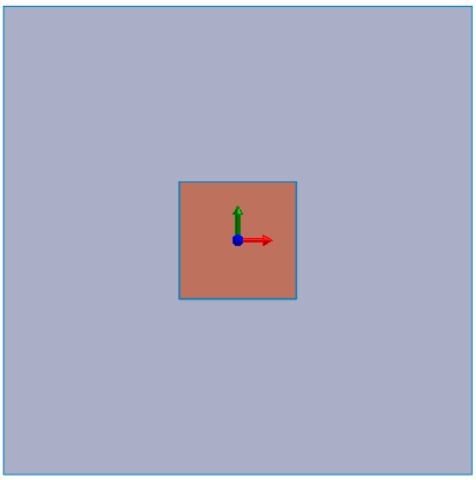


Figure 2.17: local coordinate system

and then set up the study by choosing magnetostatics study type and XY Planar problem as shown in the figure below

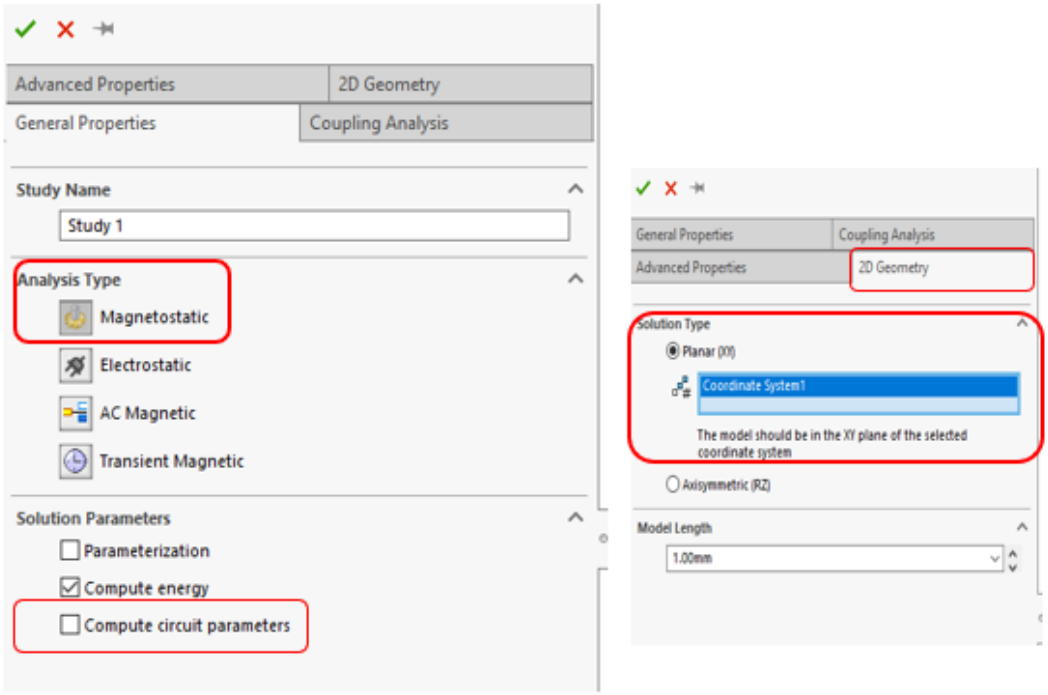


Figure 2.18: study setup

we set our materials as before and our boundary conditions, before setting the boundary condition we can shortly talk about what PDE is solved in Magnetostatics.

in equation 1.30 page 11 flux density is written in terms of the vector potential, A, as:

$$B = \nabla \times A \tag{2.2}$$

and equation 1.22 without taking displacement current into account

$$\nabla \times H = J \tag{2.3}$$

with the constitutive relation

$$B = \mu H \tag{2.4}$$

we substitute equation 2.4 into 2.3 we get

$$\nabla \times \frac{B}{\mu} = J \tag{2.5}$$

we use also equation 2.2 we get

$$\nabla \times \frac{1}{\mu} \nabla \times A = J \tag{2.6}$$

Magnetostatics solver solves equation 2.6, so we need boundary conditions on A, in most cases we need to add a large air part to our model and use Dirichlet boundary type in this type we use $A = 0$ at boundaries, and it means the flux B will not cross the boundaries

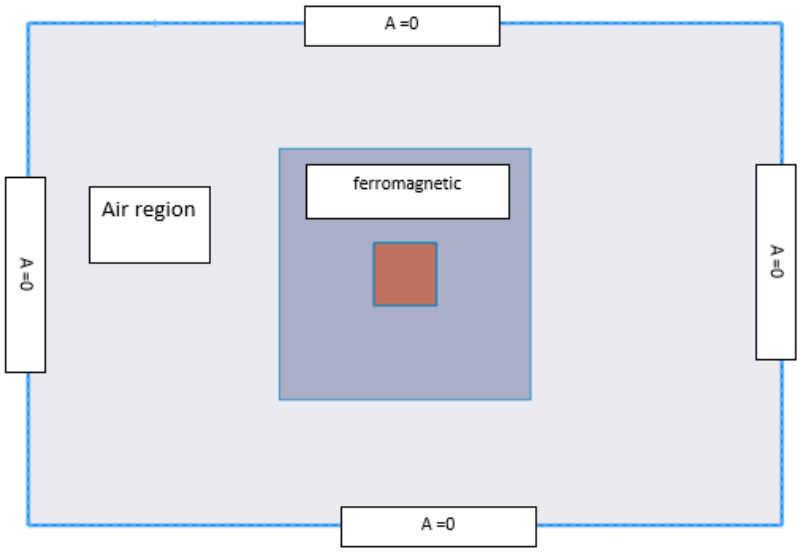


Figure 2.19: Dirichlet boundary type applied on A at air edges

going back to our simulation, we will check what shape of magnetic field is generated by a current conductor, we know that flux will be turning around the conductor if we check analytical solution

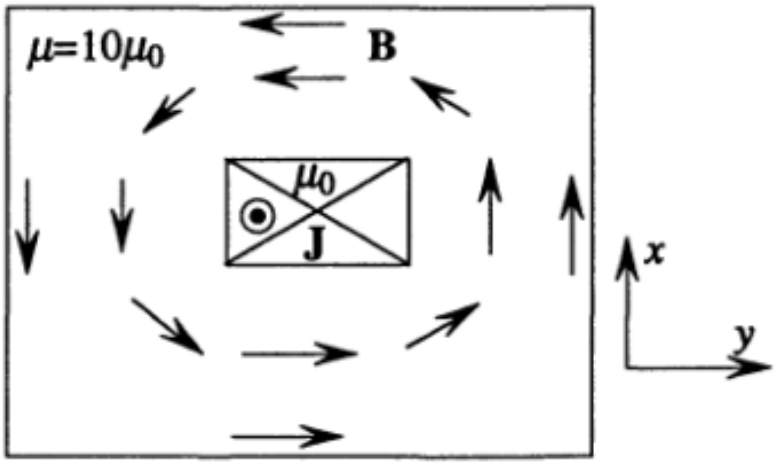


Figure 2.20: analytical solution vector plot of B

and if we increase current flux increases.

as we did before, we apply materials, copper for conductor, AISI steel for ferromagnetic and air for air region, we apply our boundary conditions

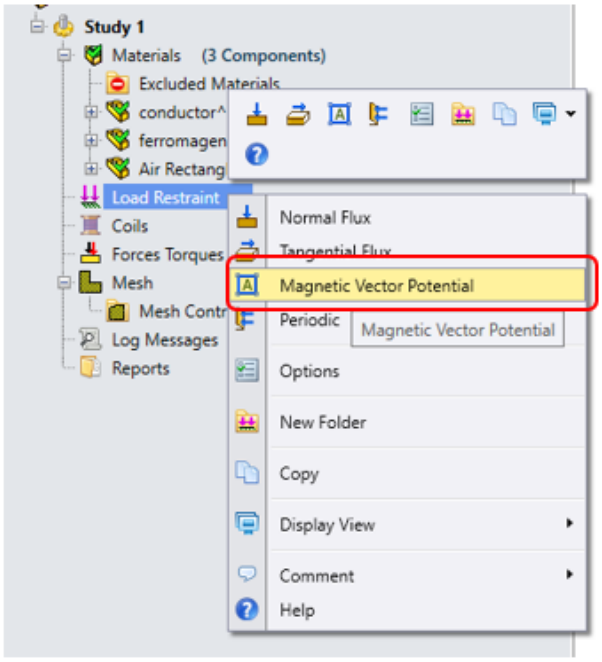


Figure 2.21: Dirichlet boundary type applied on A at air edges

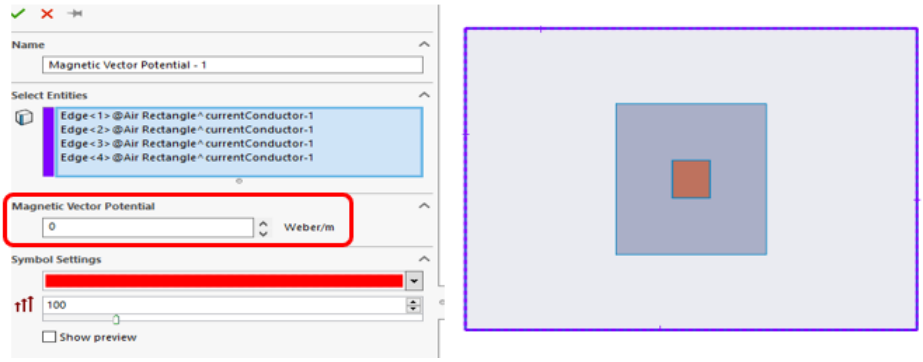


Figure 2.22: A = 0 at air edges

Since we have a conductor, we have two types that can be used, either a solid conductor or a wound conductor

for our example we will use solid coil ,with 5A net current

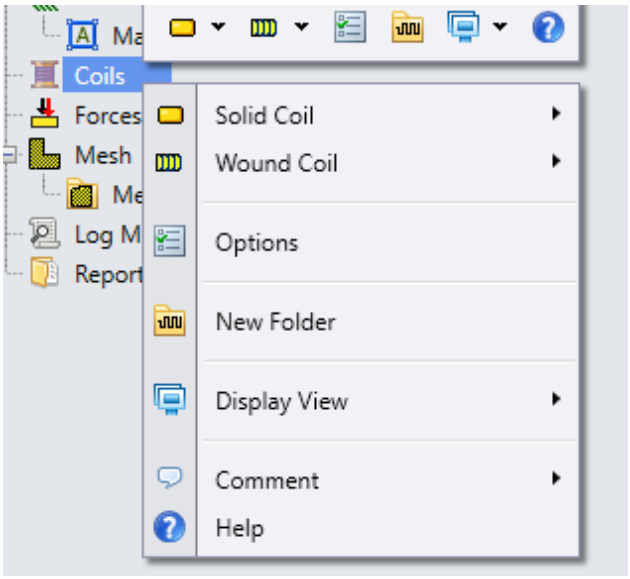


Figure 2.23: types of coils

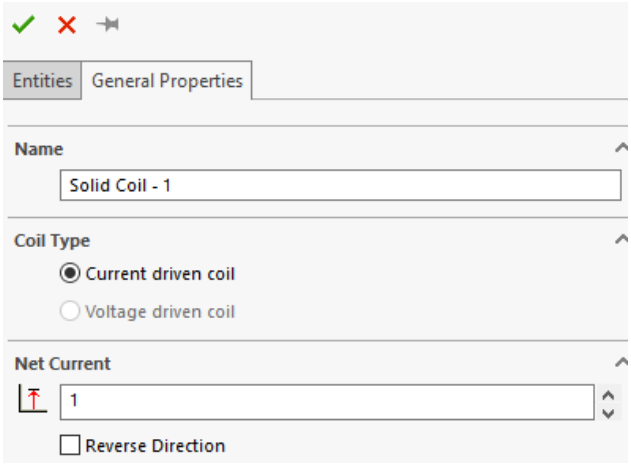


Figure 2.24: types of coils

‘ Now our study is ready to run, we can check the flux density plot

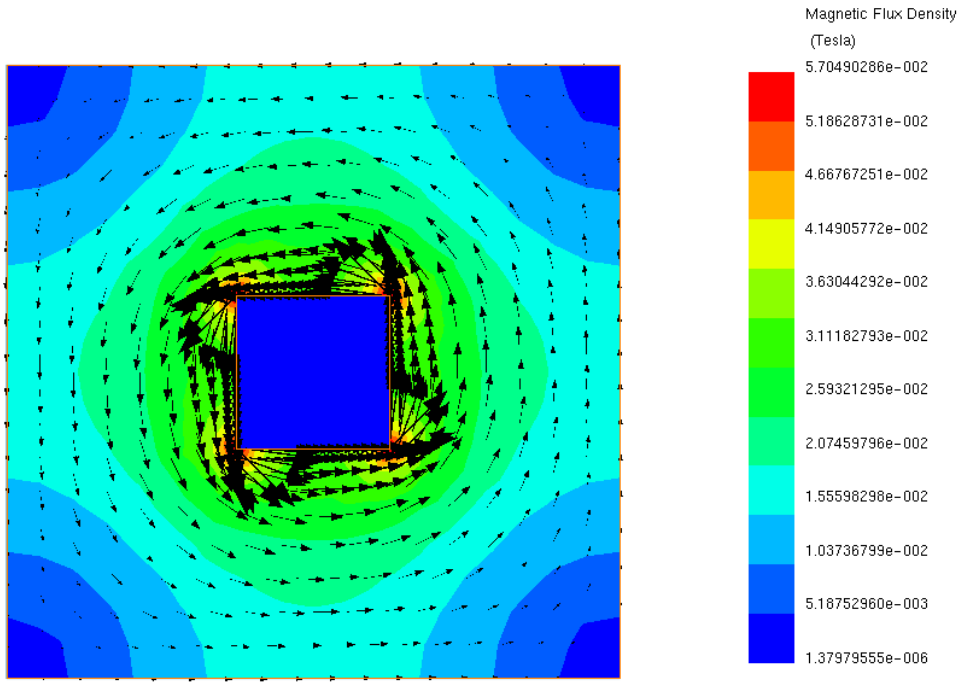


Figure 2.25: Flux density vector plot

let's check the flux variation between tow points 1 close to the conductor and another far from it

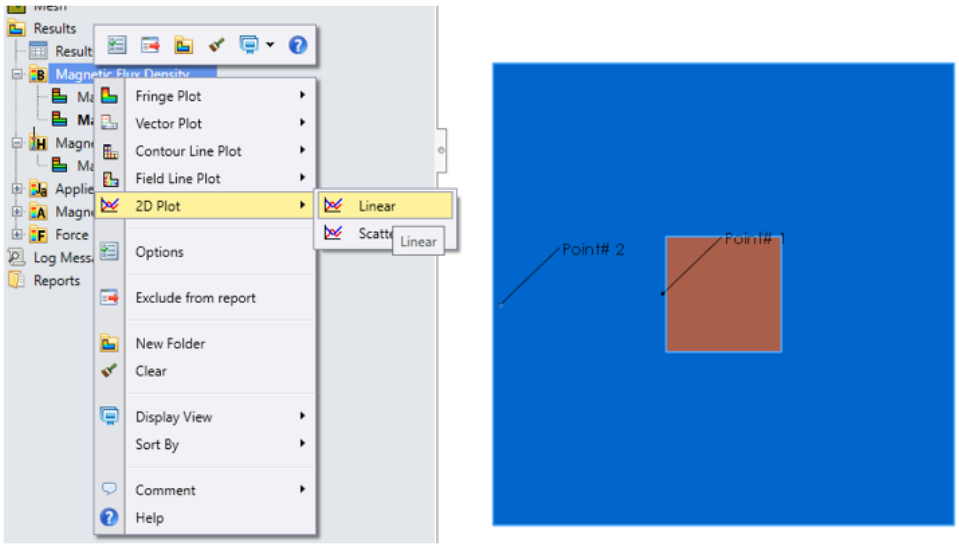


Figure 2.26: Flux density variation between tow points

if we increase current, we will get a higher flux density.

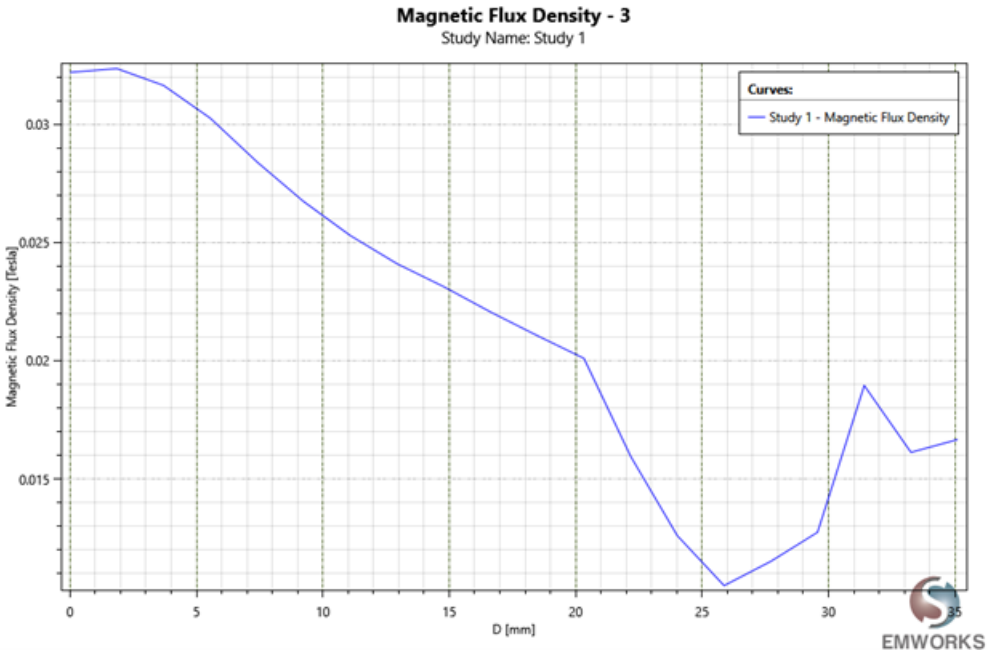


Figure 2.27: Flux density variation between tow points

2.3 Third Example : Two movable magnets

In this example we will also look into a magnetostatics problem introducing motion parametrization.

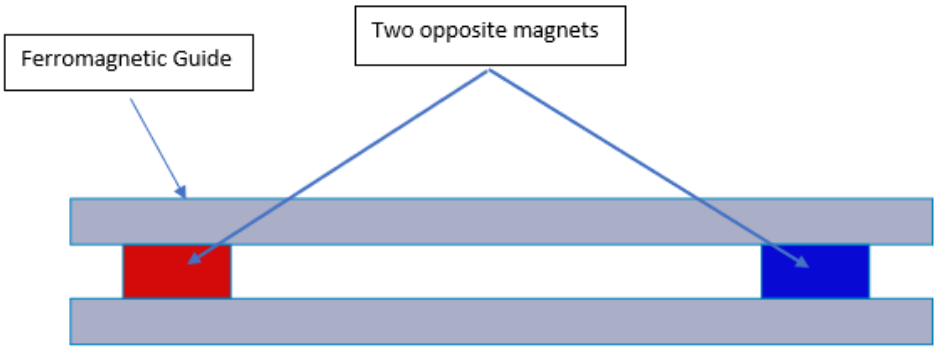


Figure 2.28: two movable magnets model

we will be interested in the force between the two magnets when they move linearly along x

as we did before we create our local coordinates system,
now we need to define the distance between the magnets that we will use it later on as our

motion variable

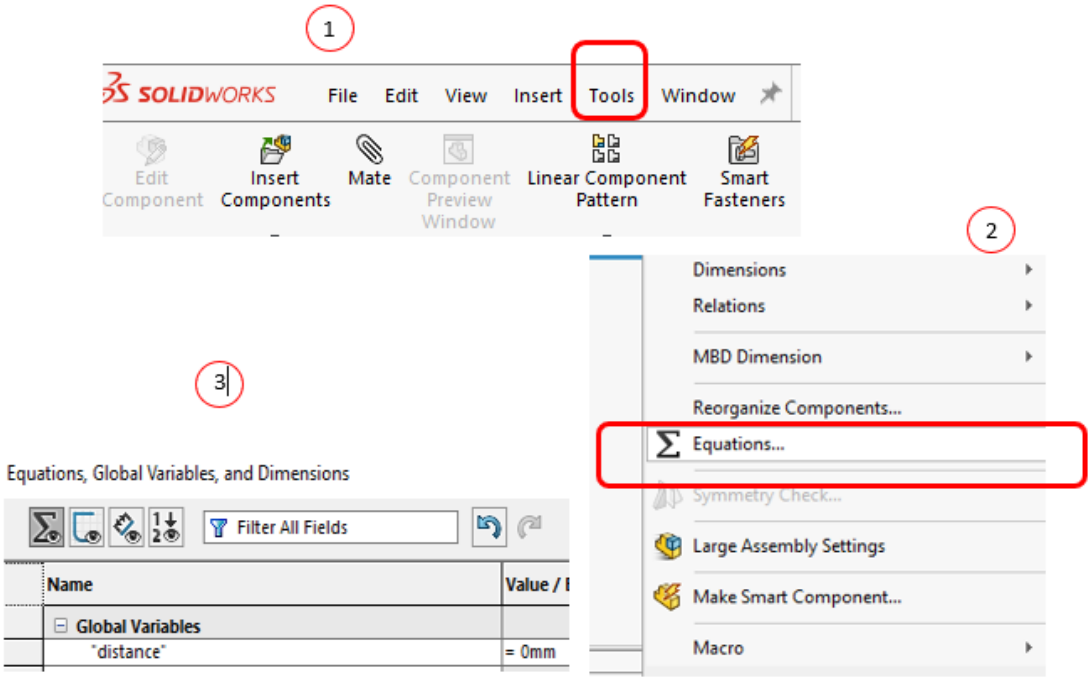


Figure 2.29: create an equation variable

and next create distance mate and link it to the equation, select the distance mate, double click and choose global variable.

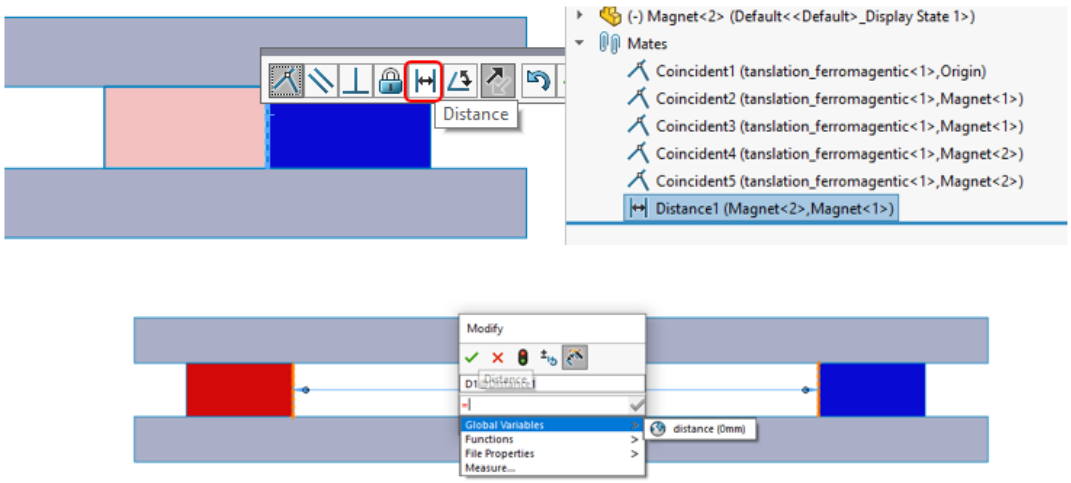


Figure 2.30: link distance mate to equation

we now create our magnetostatics study with the parametrization option and for magnets materials we can choose Neodymium magnets from permanent magnets folder

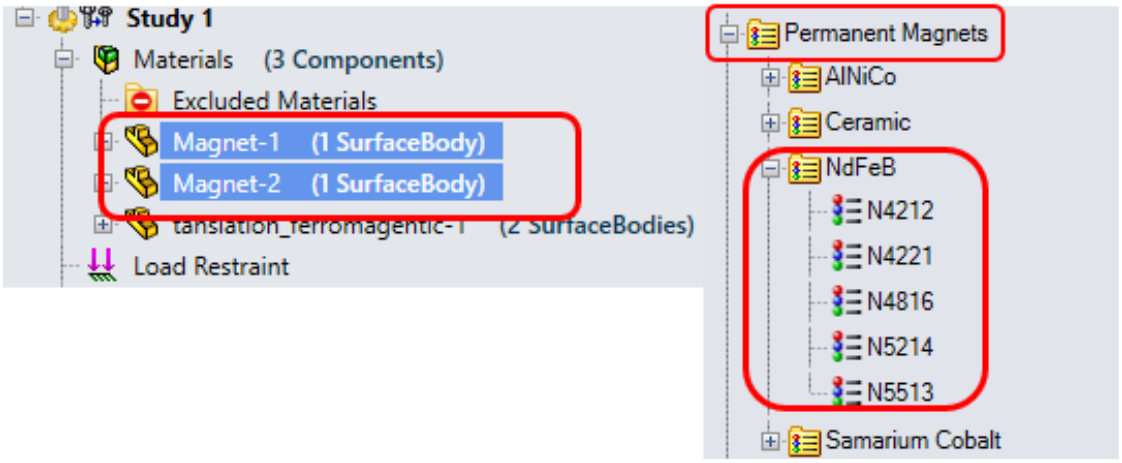


Figure 2.31: neodymium magnets

when we assign magnets to a body, we need to specify the flux direction axis, using the coercivity direction option, we chose direction type either Cartesian or cylindrical and the axis for B direction.

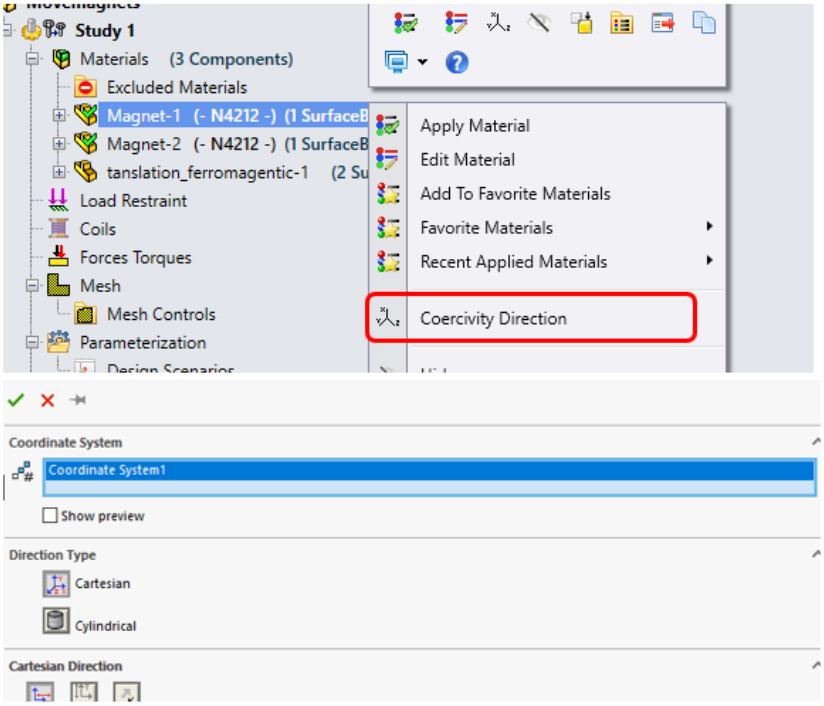


Figure 2.32: Coercivity direction

We can also preview the coercivity direction as shown below

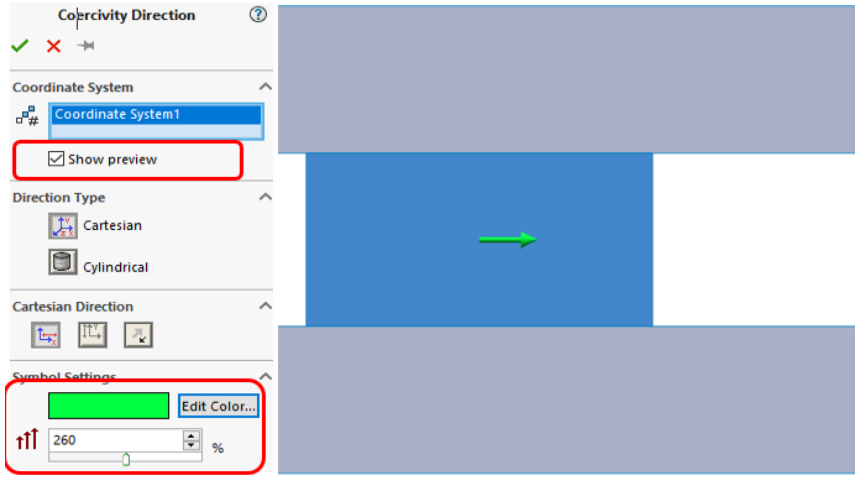


Figure 2.33: Coercivity direction preview

We then select the object where we want to compute the force, the virtual work method was chosen (this method drives force from total energy)

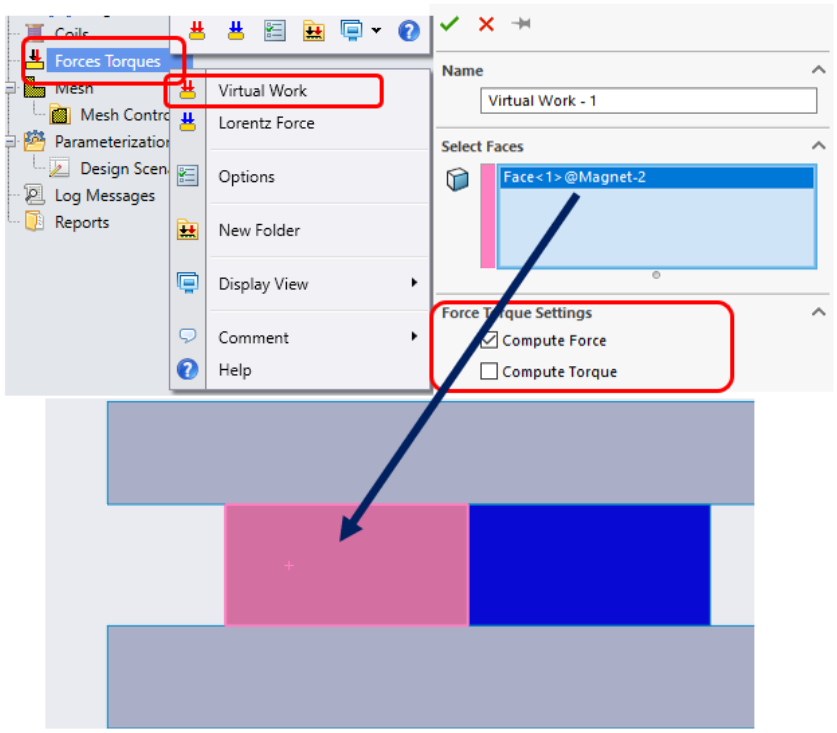


Figure 2.34: force setup

what left now is to configure parametrization, before doing so we need to load our variable that we defined before, i.e. distance by right-clicking on assembly level and selecting parameters

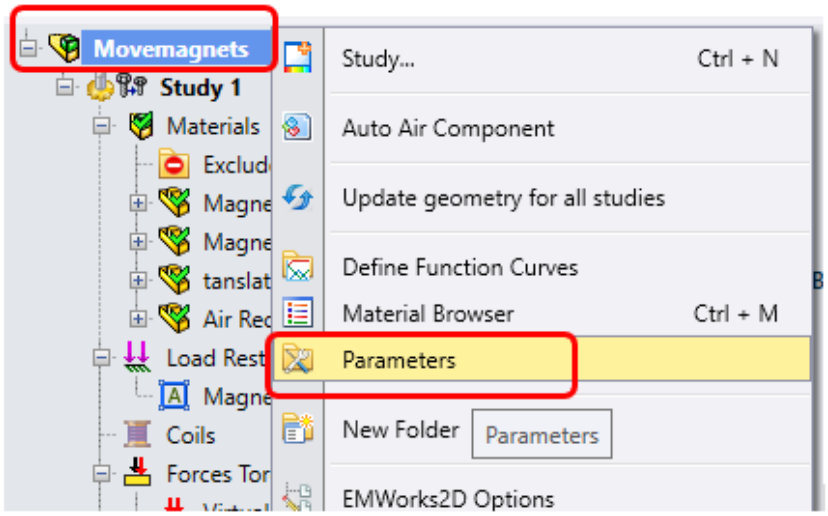


Figure 2.35: Load global variable

you end up with this window, press import global variables and then on ok

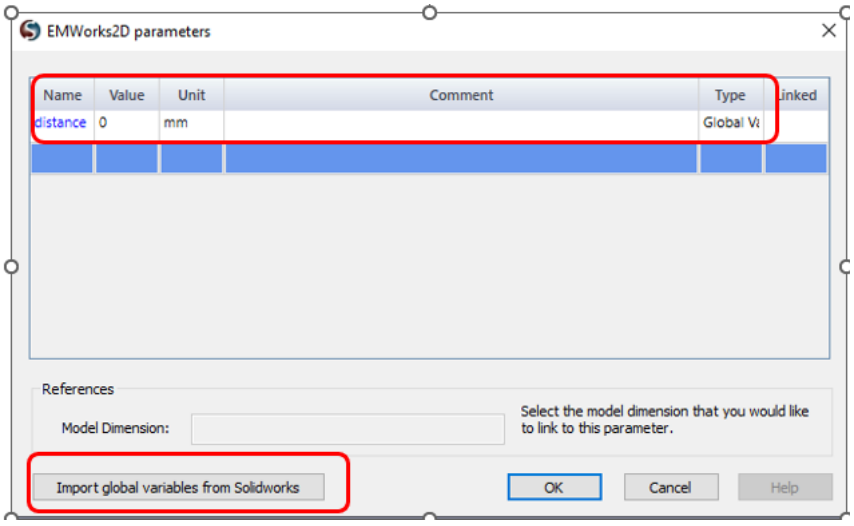


Figure 2.36: Load global variable

We can then edit parametrization and chose our distance values

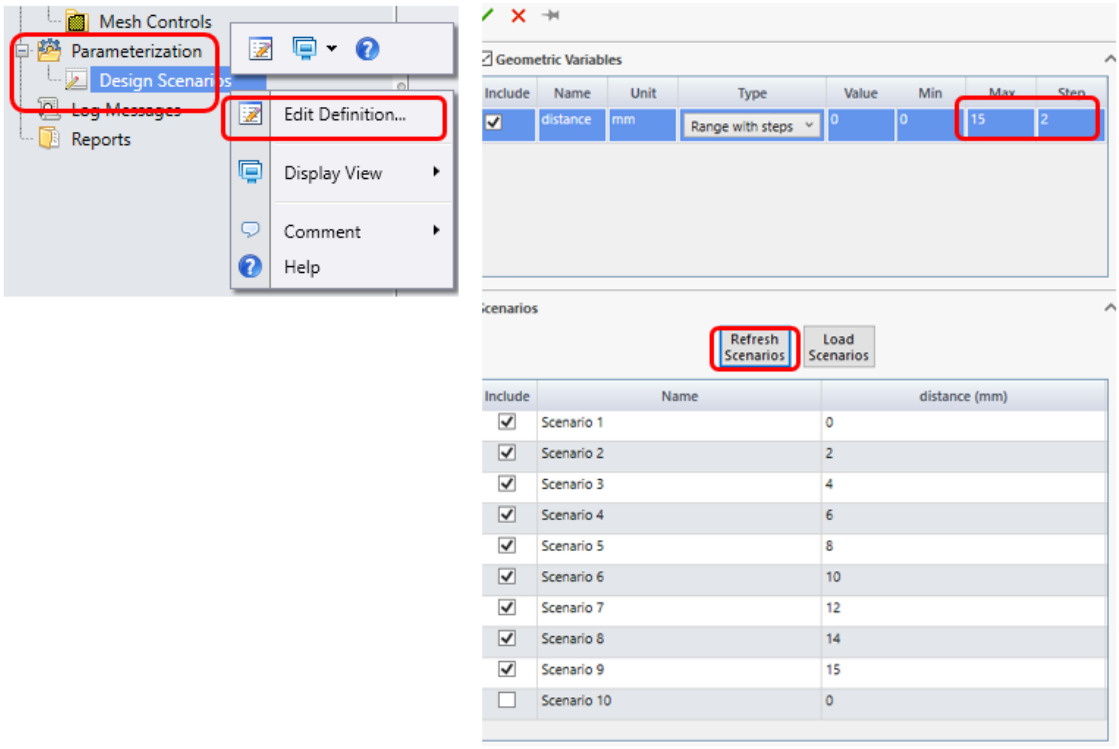


Figure 2.37: configure parametrization

And now our simulation is ready, we can run and check results
to see the plots clearly we can hide the air region and plot the flux vector plot

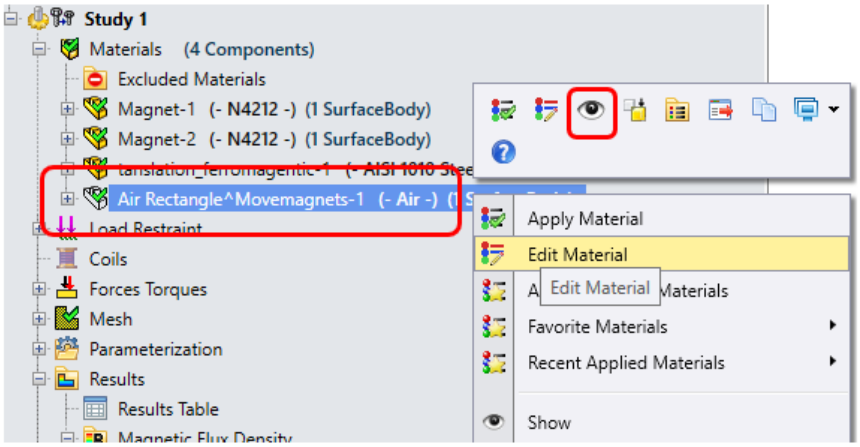


Figure 2.38: Hide Air

since the magnets are opposite, we expect a repel force to be strong if magnets are close and become weaker and weaker when magnets moves away

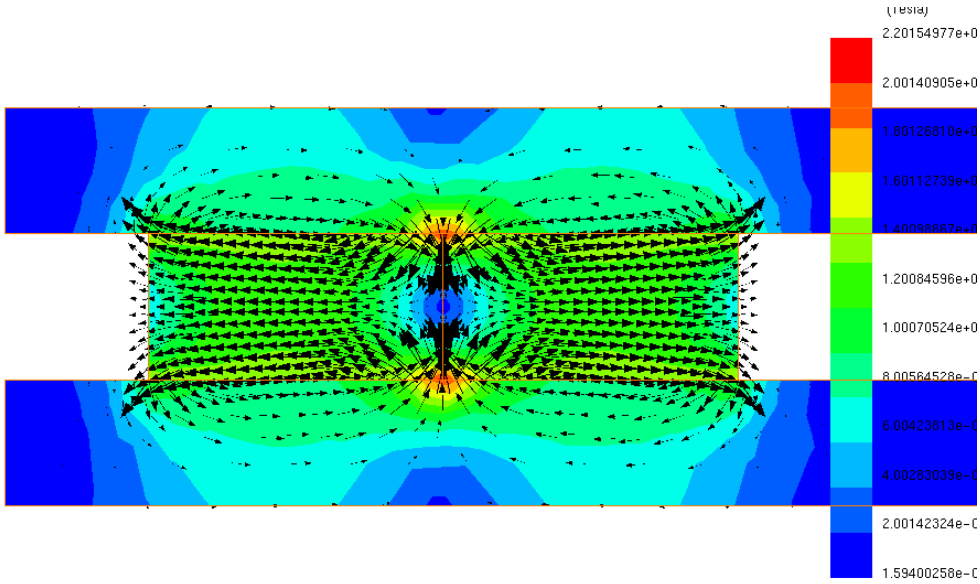


Figure 2.39: flux density vector plot at scenario 1 (Distance =0)

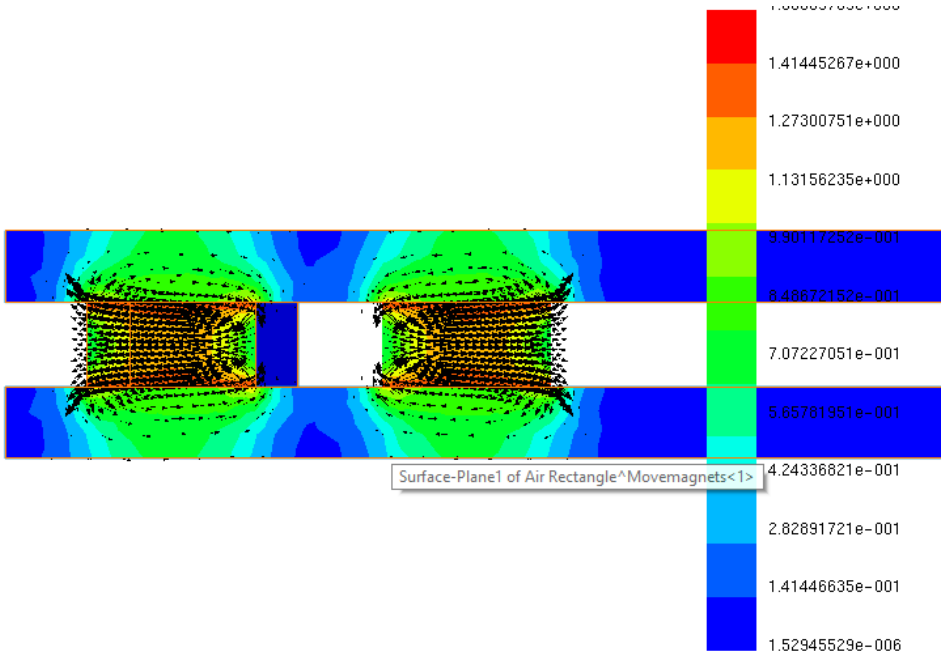


Figure 2.40: flux density vector plot at scenario 9 (Distance =15mm)

let's check how force changes with distance, we open results table
and we get this result, and we can see that force applied to the selected magnet is negative

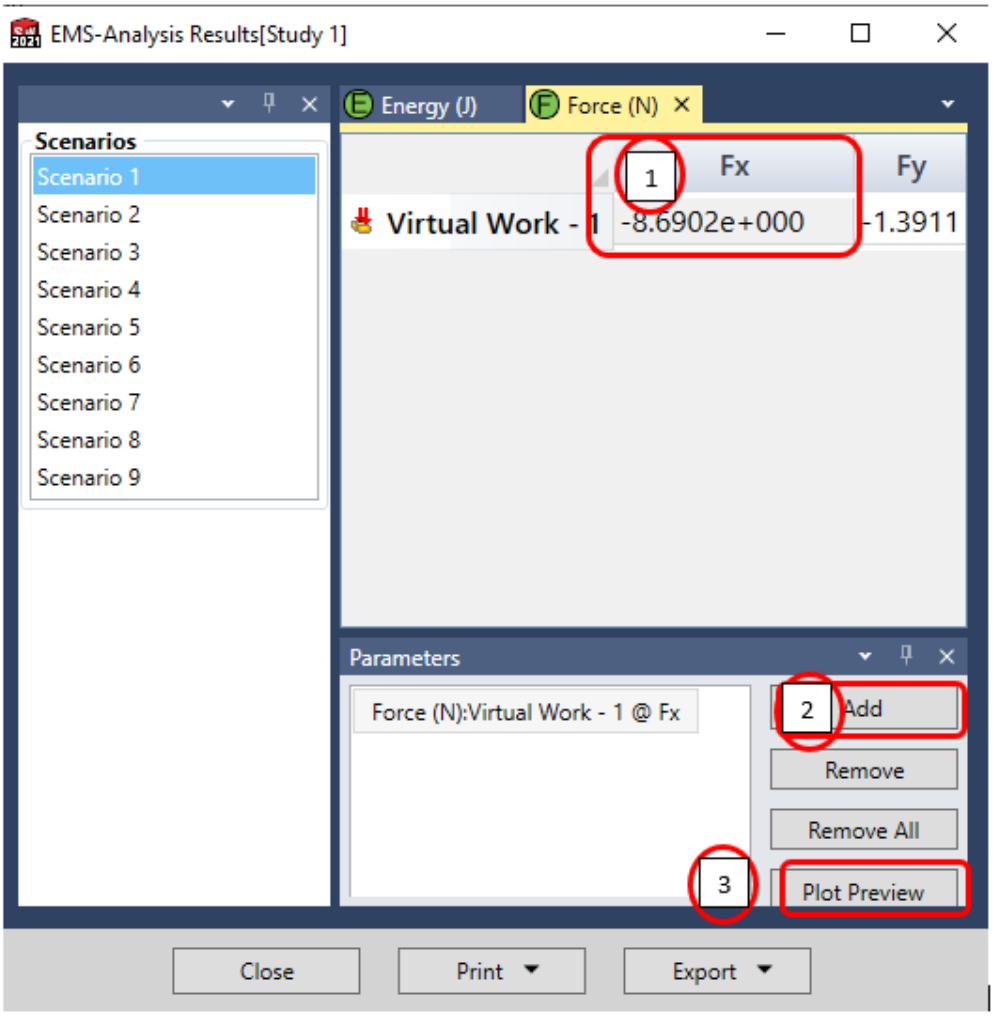


Figure 2.41: plotting force curve

in x direction and its absolute value decreases from scenario to scenario

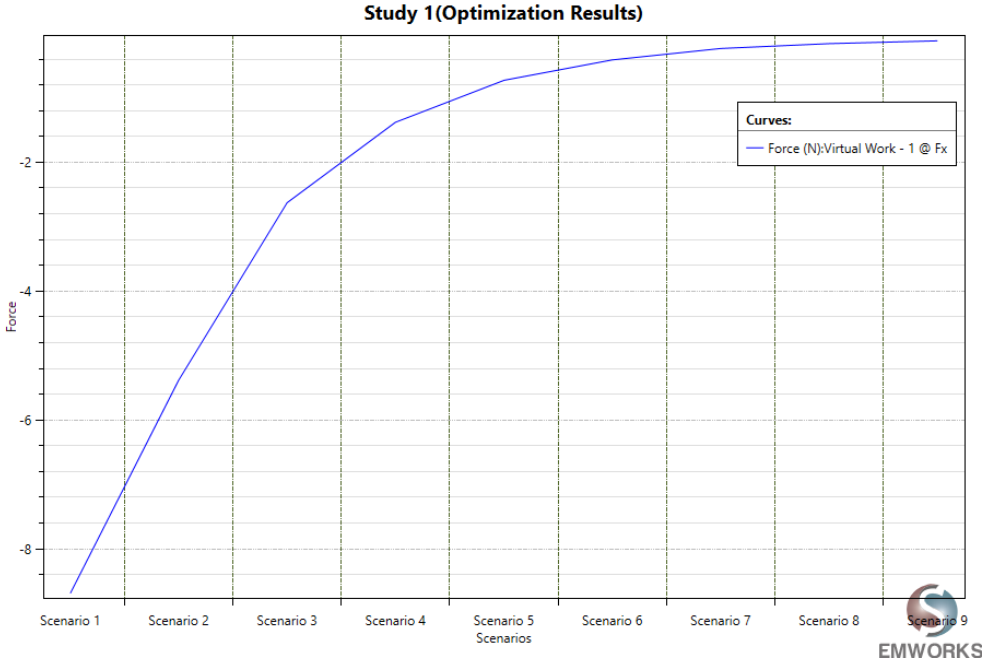


Figure 2.42: force curve

2.4 fourth Example : Transient Magnetic simulation of a 3D Electric Motor

In this example, we will simulate a 3D motor and determine its characteristics under No load and on load conditions, such as torque and power loss ...

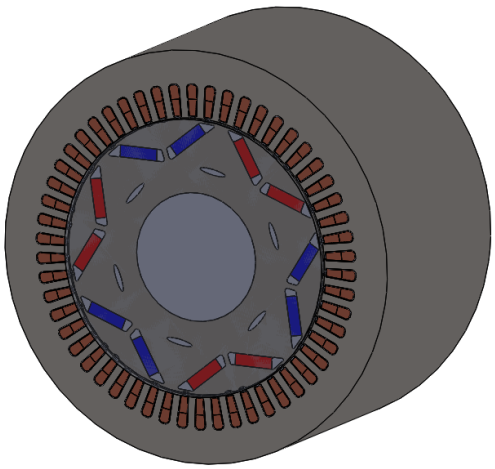


Figure 2.43: Tesla Model 3 Motor

in 3D Generally we don't need to define a local coordinates system, however in this model the direction of flux in each magnet can't be assigned using a global coordinates' system, so we need to define a local coordinates system for each magnet.

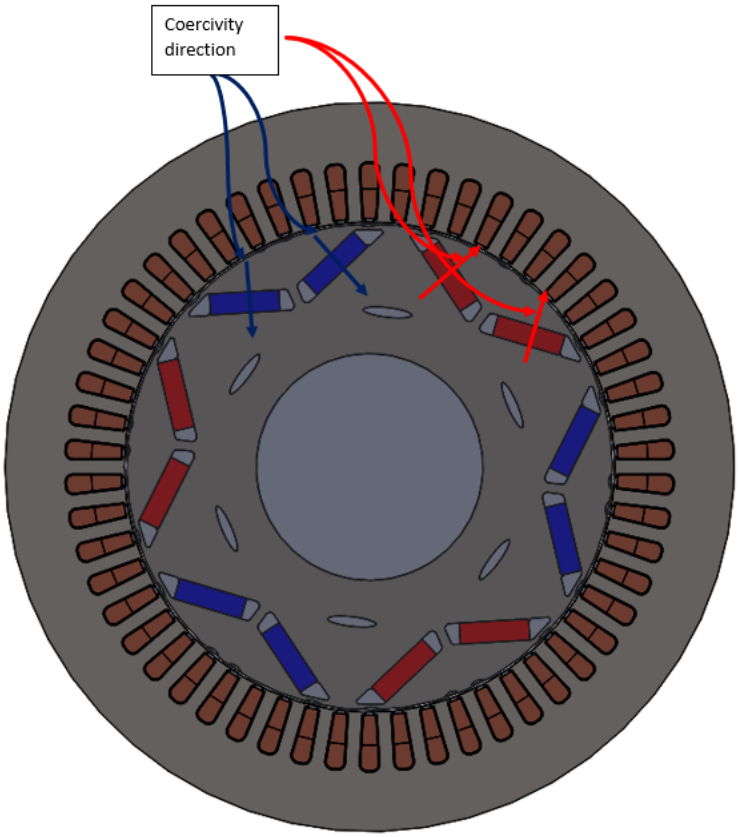


Figure 2.44: Direction of flux needs a local coordinates system to be defined properly

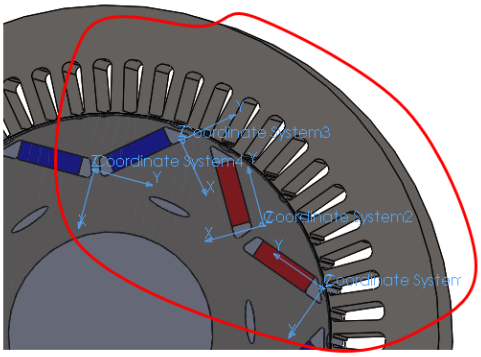


Figure 2.45: Local coordinates system

2.4.1 The band Component

When we have a moving part in our model, we need to use what is called a band, the Band should enclose all the moving parts, and the remaining bodies should be fixed

If we already have our model designed, we can simply add a new part (Insert ⟶ Component ⟶ Part) and the band should enclose the rotor as shown in this figure

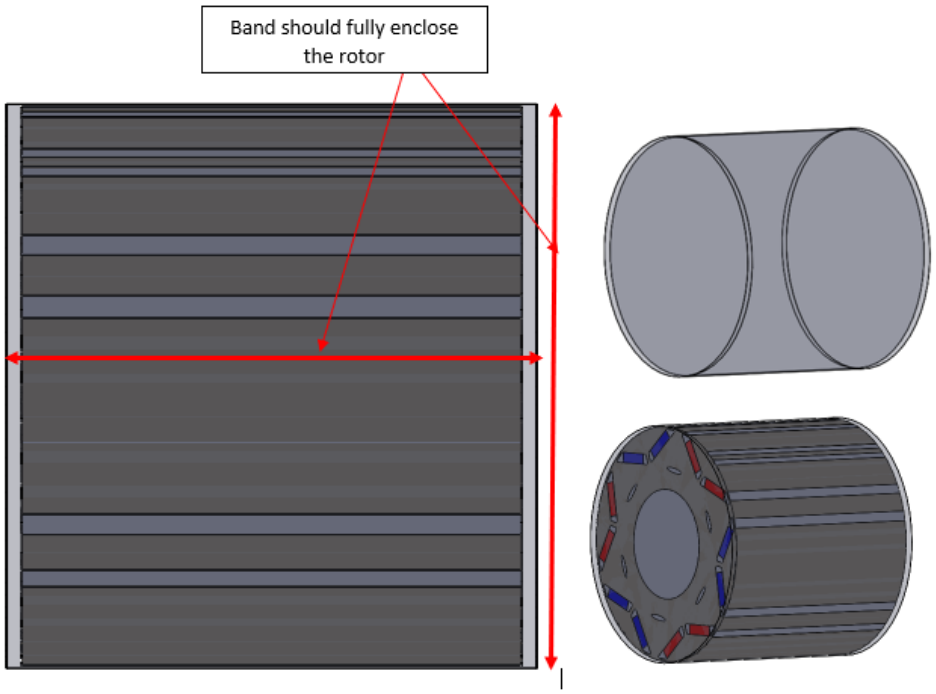


Figure 2.46: the Band component

a Final step is to add a cavity on the band and check for interference (inside the band part, Insert ⟶ Features ⟶ Cavity)

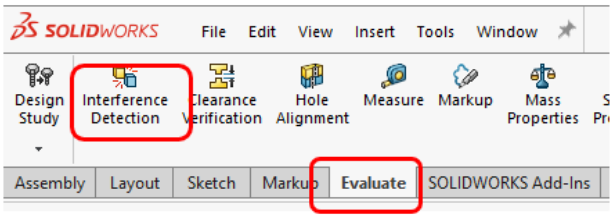


Figure 2.47: Interference check

2.4.2 Motion setup

We choose the rotation direction and the speed of rotation

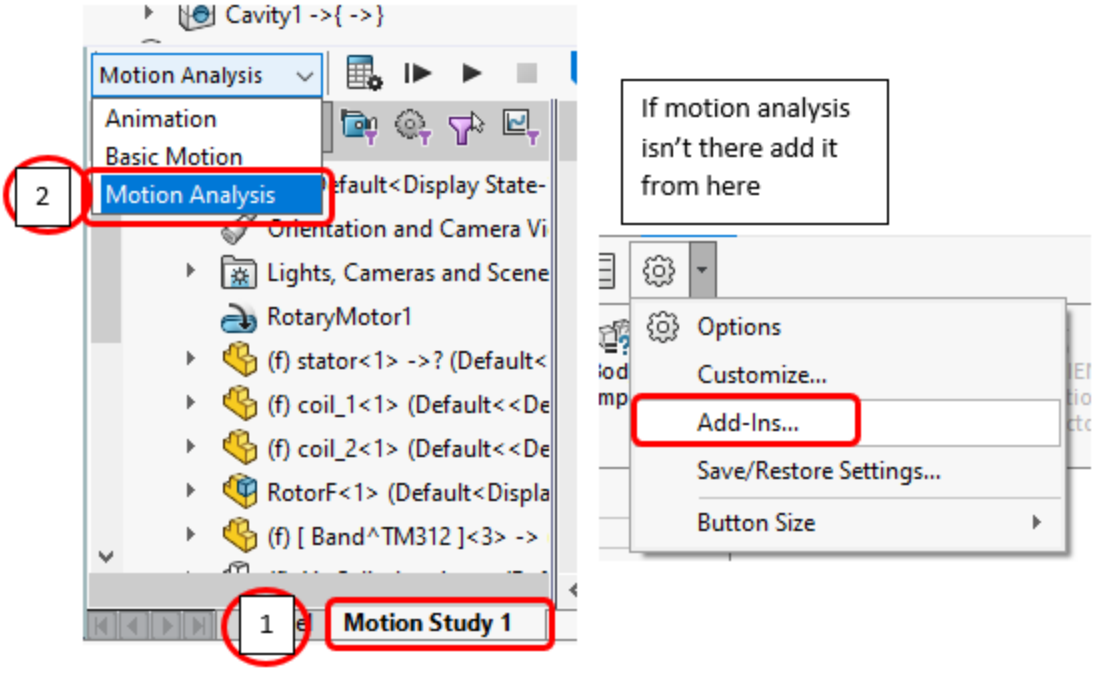


Figure 2.48: Motion analysis option

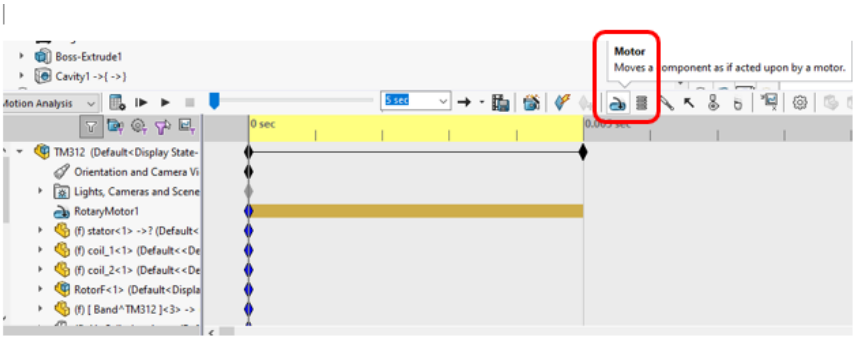


Figure 2.49: adding a motor

now we have to choose how much in each time step the rotor rotates by choosing the end time and time step of simulation

2.4.3 Creating a Transient magnetic study

we select transient magnetic study and then in the coupling analysis we select motion coupling, the time step and end time will be set automatically after selecting motion according to the number of frames per second and end time that we have chosen previously.

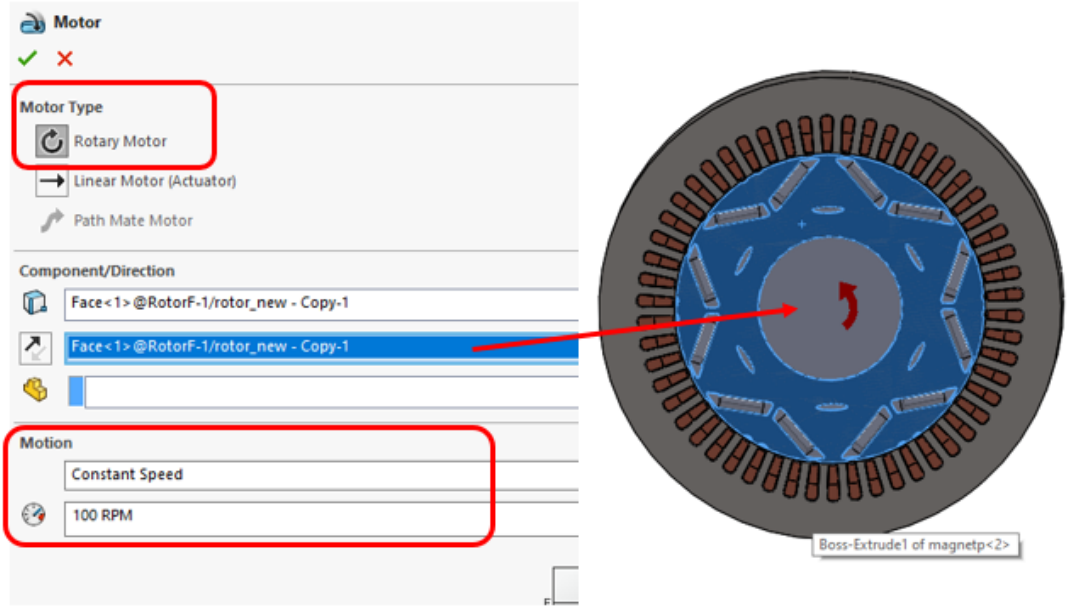


Figure 2.50: speed configuration

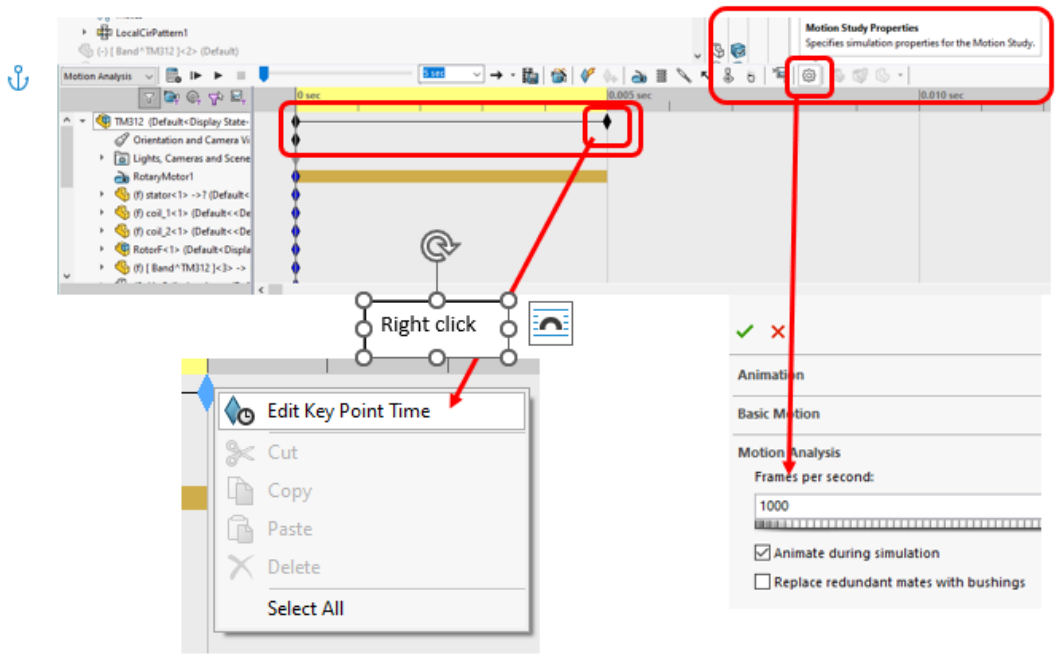


Figure 2.51: Simulation time configuration

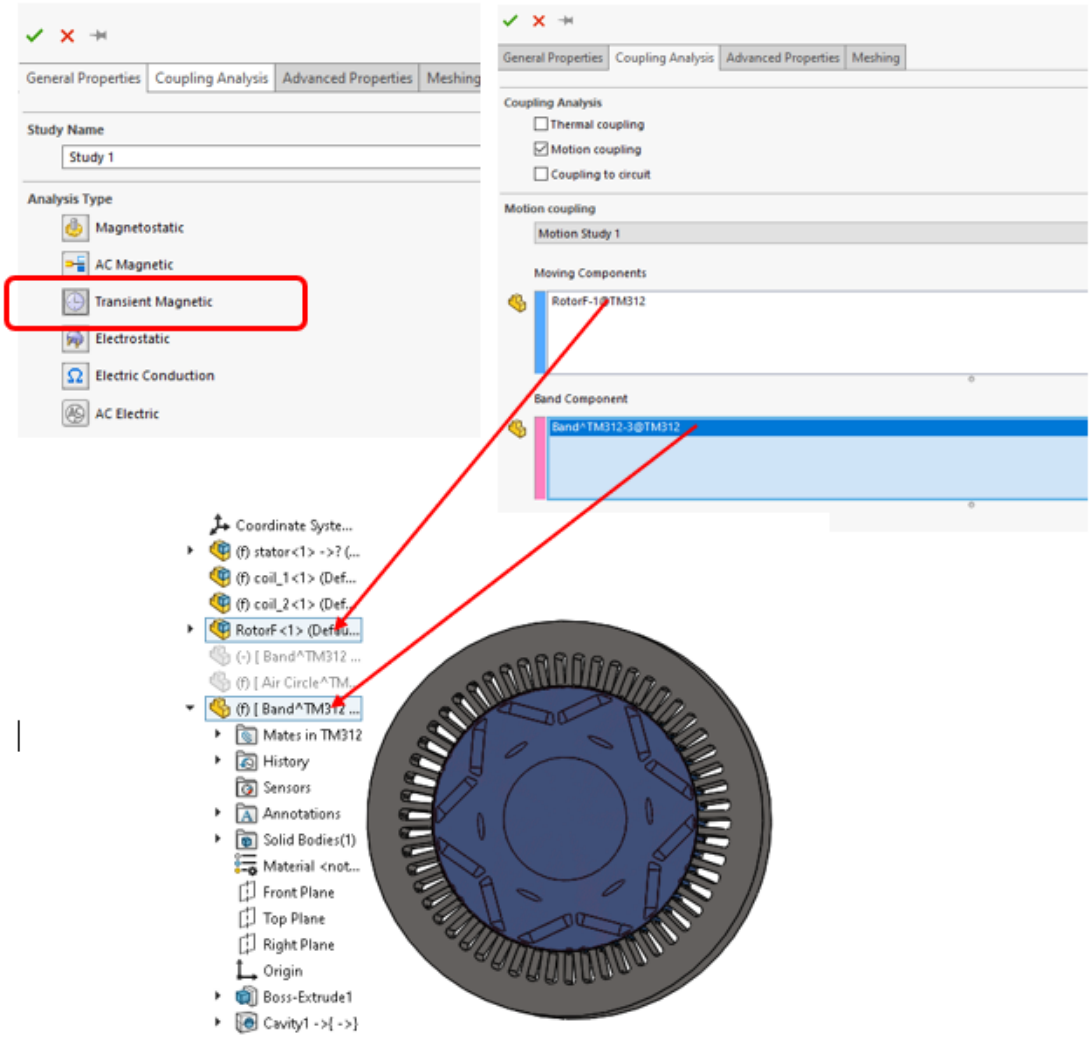


Figure 2.52: study creation

We need now to add a big air part, that we use to apply our boundary conditions, if the air is big enough we can assume that the magnetic field is very low at the edges, by applying the Dirichlet boundary condition at the surfaces of the air region(A =0), in EMS the boundary condition is applied automatically, and we don't need to specify it, we can create an automatic air region as shown below

next we apply the materials and where we want to compute force or torque,

2.4.4 Winding configuration

There are different types of winding configurations, Concentric winding ,distributed winding...single layer or double layer winding, partial slot and full slot windings ...

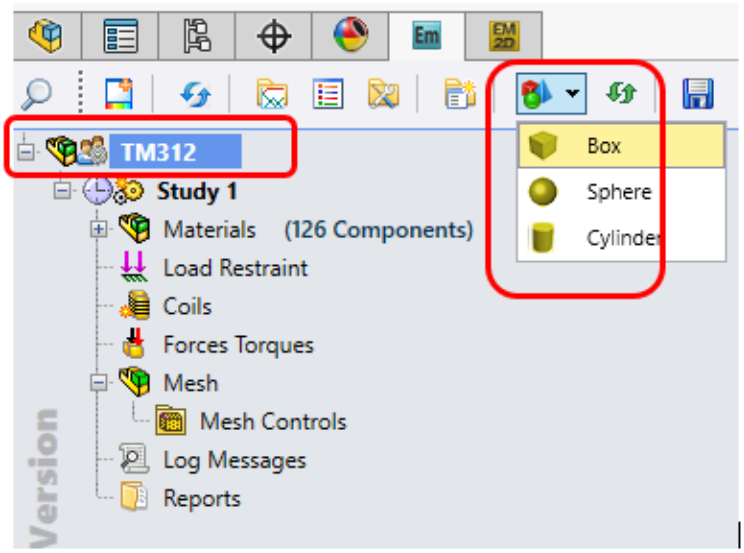


Figure 2.53: automatic air creation

in this example, the winding is a double layer distributed winding, we click on coils, and we set up our 3 phases as shown below

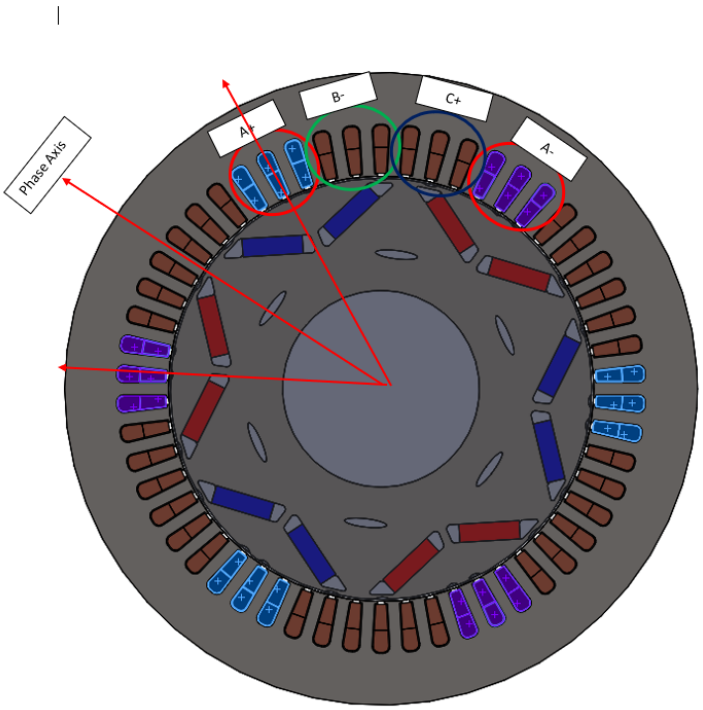


Figure 2.54: Phases Configuration

in each face, we need to select from where current enters and exits

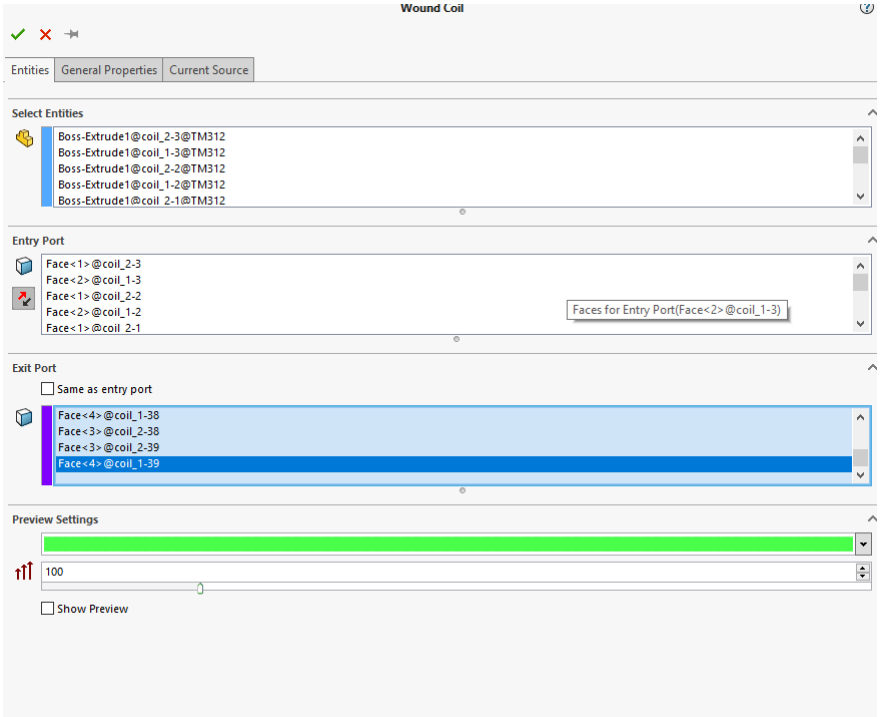


Figure 2.55: entry and exit of current

We will perform a no load simulation, we set the phase to be a current-driven winding and current magnitude to 0A, which means open circuit simulation, we will check the induced voltage waveform, we can preview the direction of current to make sure that everything is fine

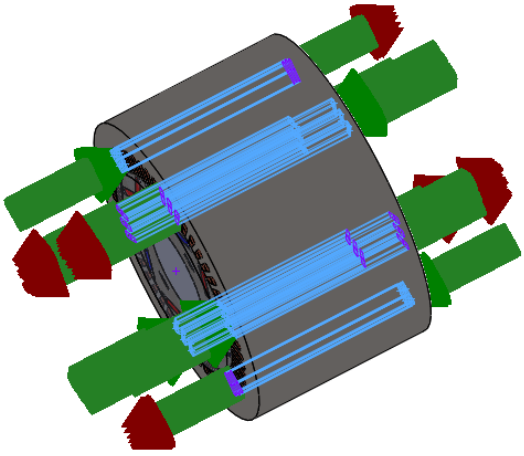


Figure 2.56: current direction preview

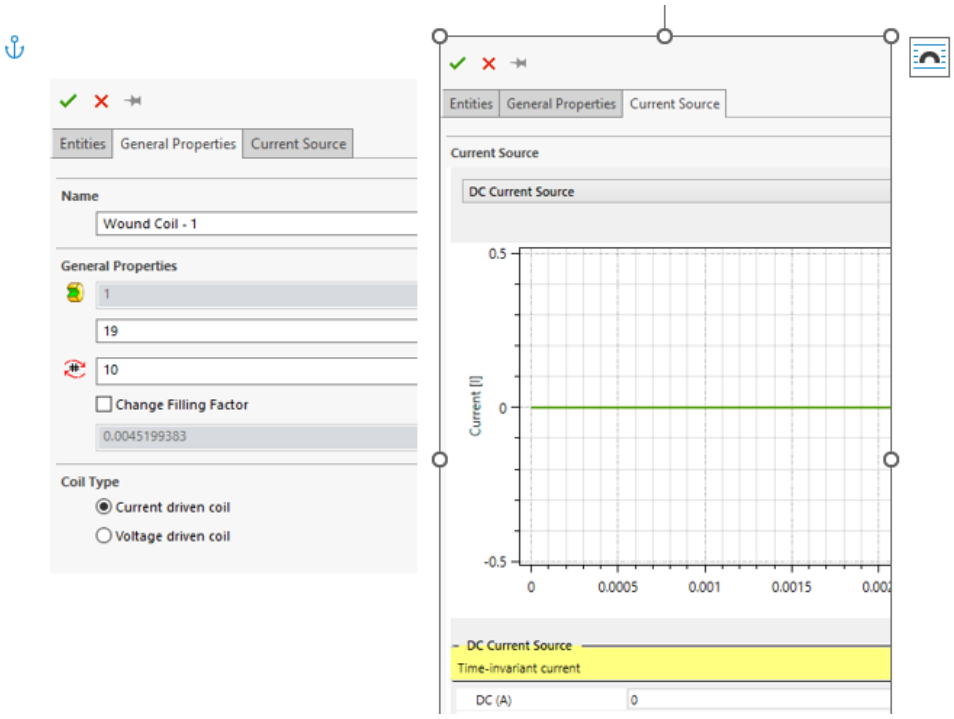


Figure 2.57: Number of turns and current excitation

2.4.5 Simulation

Just as before, we can check flux linkage (flux passing through coils) and induced voltage(or back emf ,induced due to flux change inside coils) from the results table.

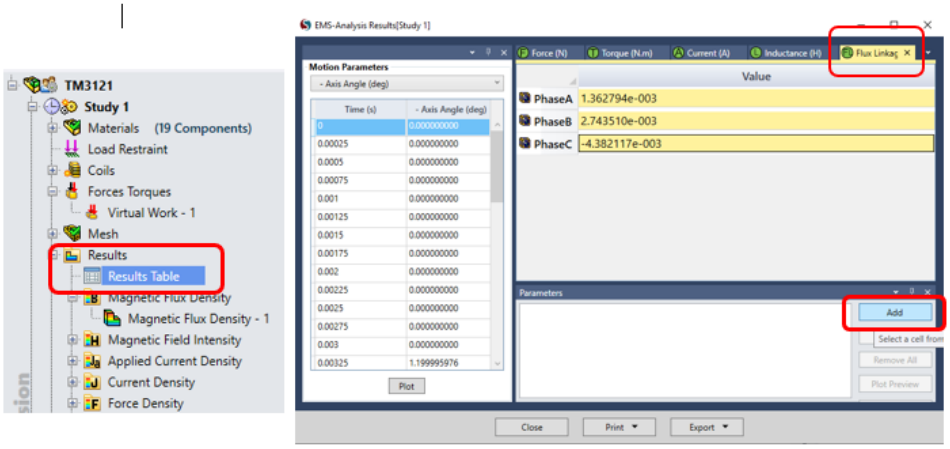


Figure 2.58: Results

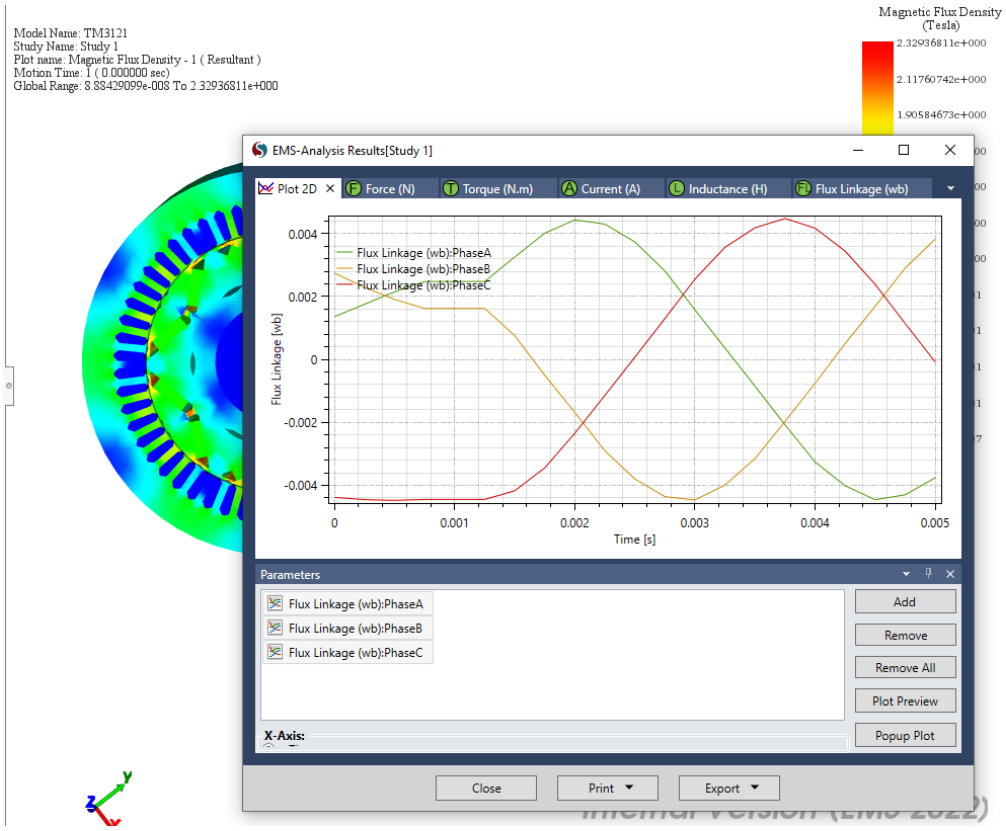


Figure 2.59: Flux linkage waveform

We can see clearly that the flux linkage is close to a perfect sine wave, which means that the motor is perfectly designed

We can check the torque(Tz in z direction) caused by the magnetic interaction between magnets and stator ferromagnetic, called cogging torque.

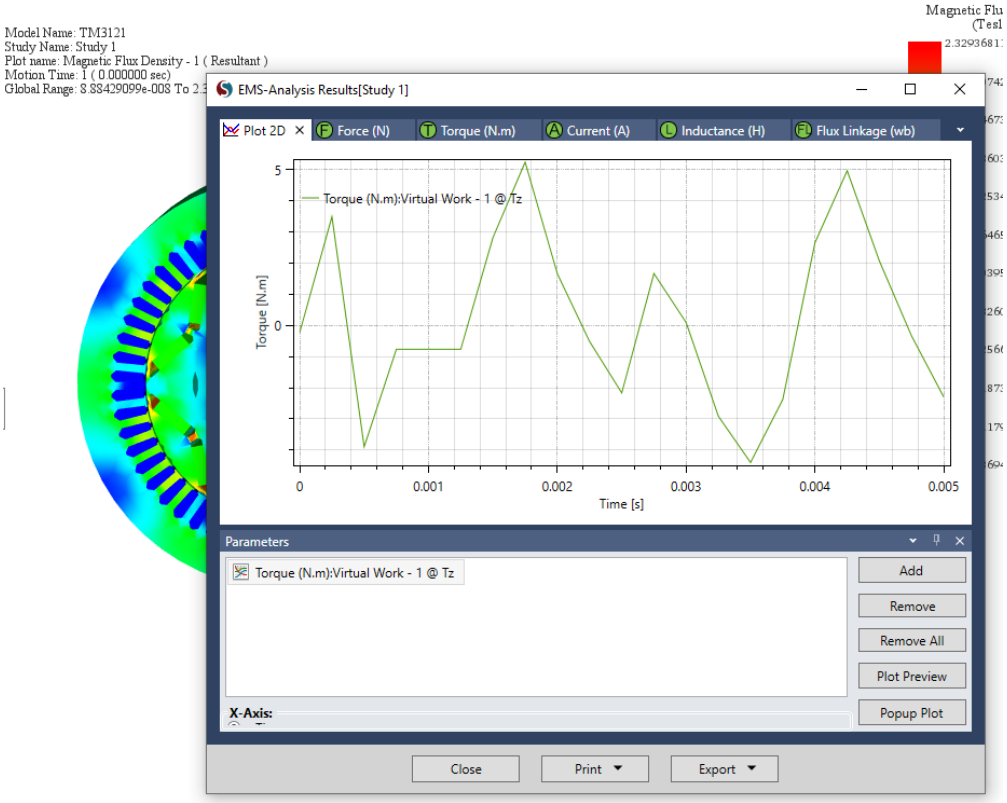


Figure 2.60: Cogging torque

for on load simulation, we use current driven, set the current to sine wave with 200Hz frequency and 970A RMS current

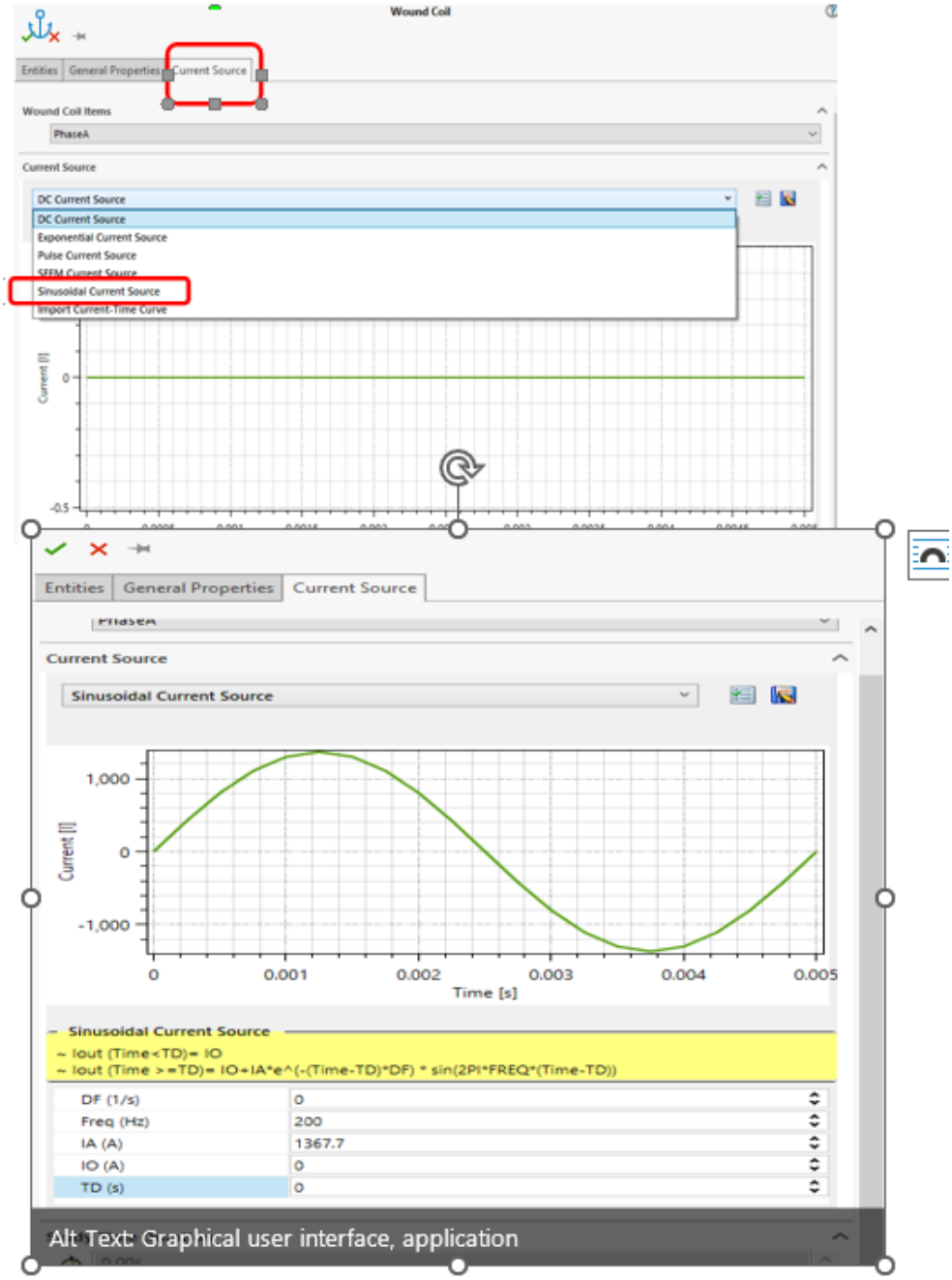


Figure 2.61: on load current excitation wavefrom

in an on load simulation for a permanent magnet motor, to get maximum torque, we talk about current advance angle, and direct and quadrature inductances, for example we can set the phase A axis(see the below figure) to be coincident with the rotor quadrature axis($\theta = 0$

and use 56deg advance angle for the 3 phases currents

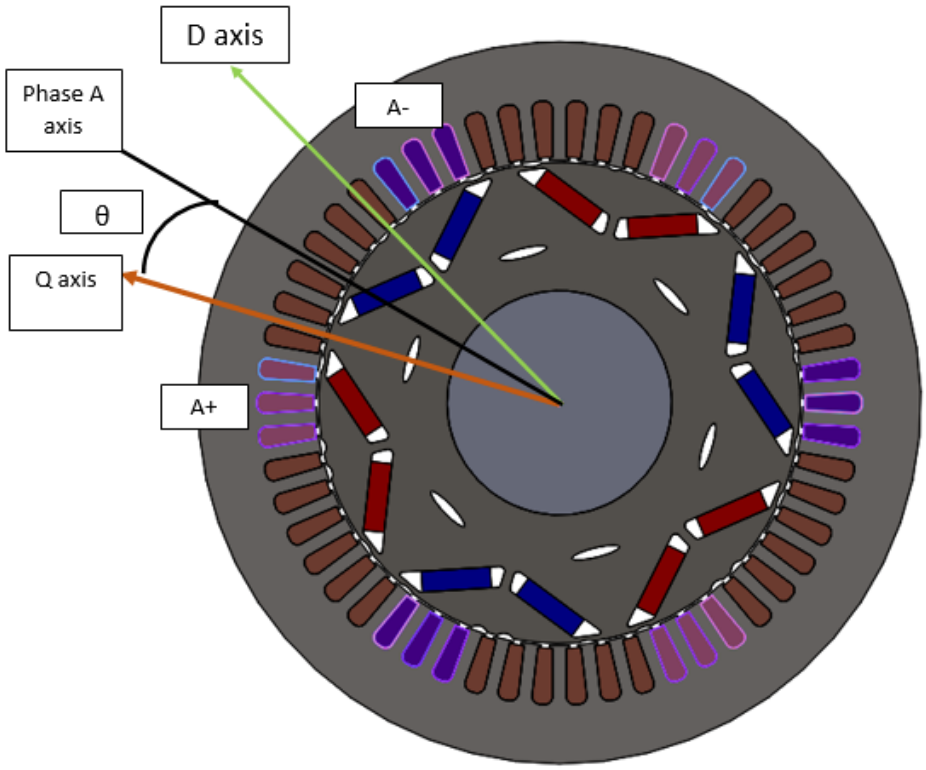


Figure 2.62: Phase axis and direct axis(d axis)

the angle was regulated manually and a new motion study is created

in re-simulating this example, you likely need to try things out to get the proper results, sometimes you need to invert motion direction or current phasing angle between two phases, It's worthwhile to play with the setup to get used to those kinds of things.

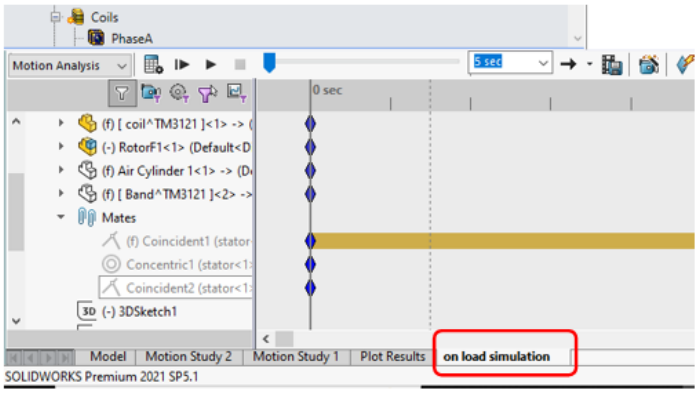


Figure 2.63: phase axis alignment with the quadrature axis

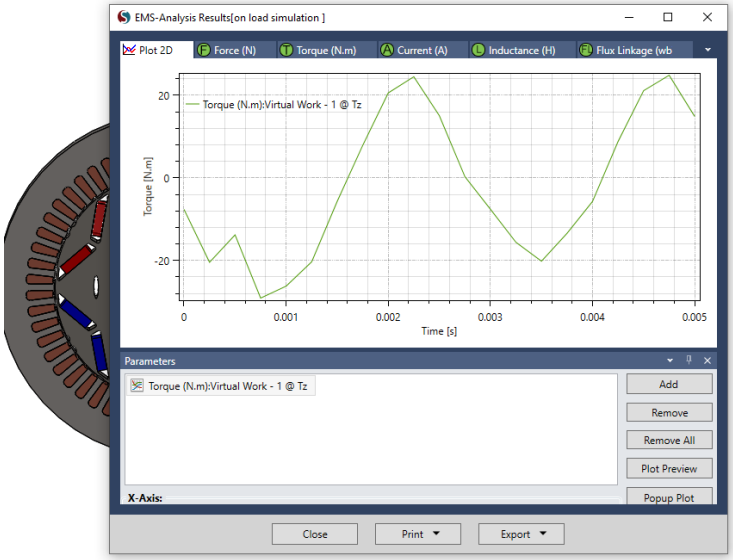


Figure 2.64: Torque

it appears that the torque curve crosses the time axis and becomes 0 at some time steps ,which means we need to invert the direction of motion so the virtual motion and flux rotates in the same direction

2.5 Fifth example :From 3D to 2D

In some cases, our problem has symmetry, in electromagnetic that means both geometric and electric symmetries (or invariant according to an axis), what that means, is the flux density

for example doesn't depend on the z component in a planar symmetry ,and depends only on x, y components or equivalently the magnetic vector potential has only the z component since B is the curl of A, in those cases we can simplify the problem and instead of simulating the whole 3D model we only simulate a planar section from it, and take the length of the model into account to extend the results.
going back to our previous 3D example, we can see that the geometry and phases displacement is the same in the X, Y plan, So instead we will now simulate a 2D section.

$$B = \boldsymbol{\nabla} \times A \tag{2.7}$$

$$B_x\vec{i} + B_y\vec{j} + B_z\vec{k} = (\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z})\vec{i} - (\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z})\vec{j} + (\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y})\vec{k} \tag{2.8}$$

$$B_z = (\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}) = 0 \tag{2.9}$$

2.9 means

$$\frac{\partial A_y}{\partial x} = \frac{\partial A_x}{\partial y}$$

meaning that either A_x and A_y are constant or 0

we can assume that they are 0 and search for the corresponding z component, since they are not involved in computing B

2.5.1 2D simplification

to go from 3D to 2D, we don't need to recreate a surface geometry we can use a feature in EMWorks2D called 2D simplification, before using it we can get rid of some parts that we can create them simply later on, (the air, the band) and we need to insert a local coordinates system

ensure that the XY axis are in the right place and use 2D simplification

once simplification is done, we end up with this model

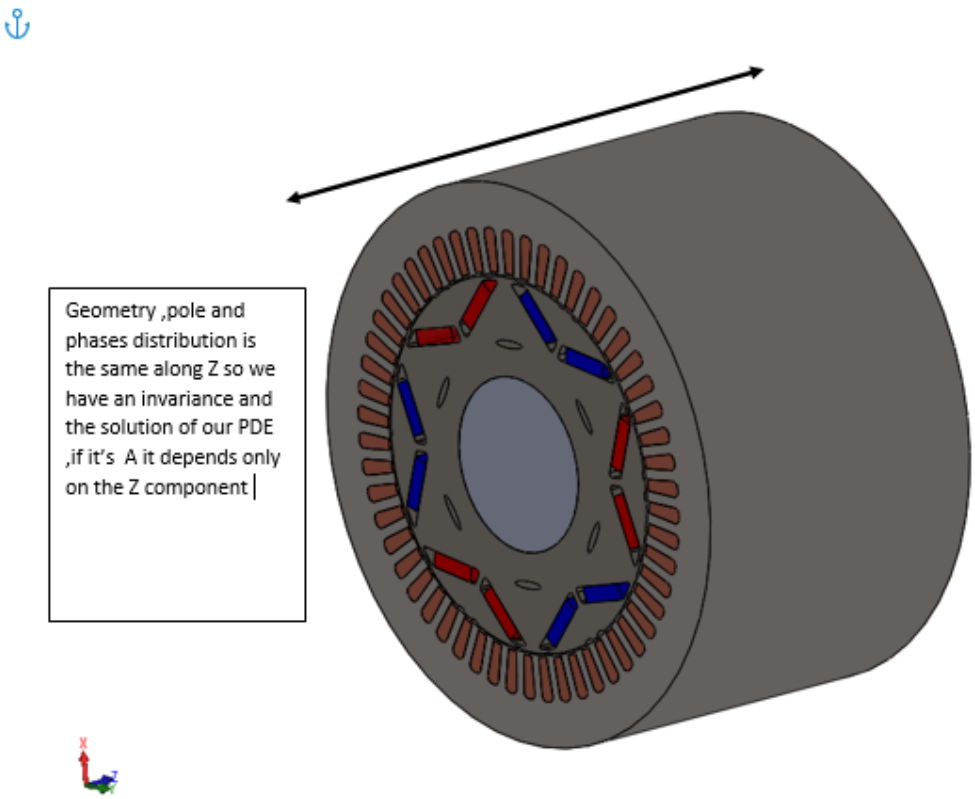


Figure 2.65: XY symmetric model

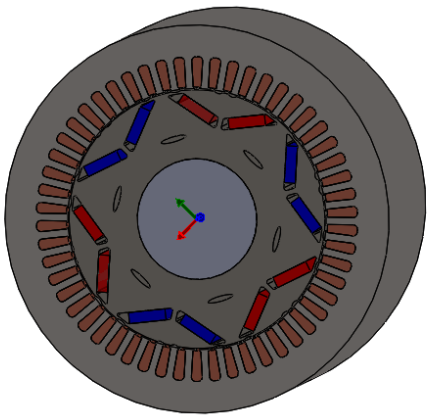


Figure 2.66: Local coordinates system

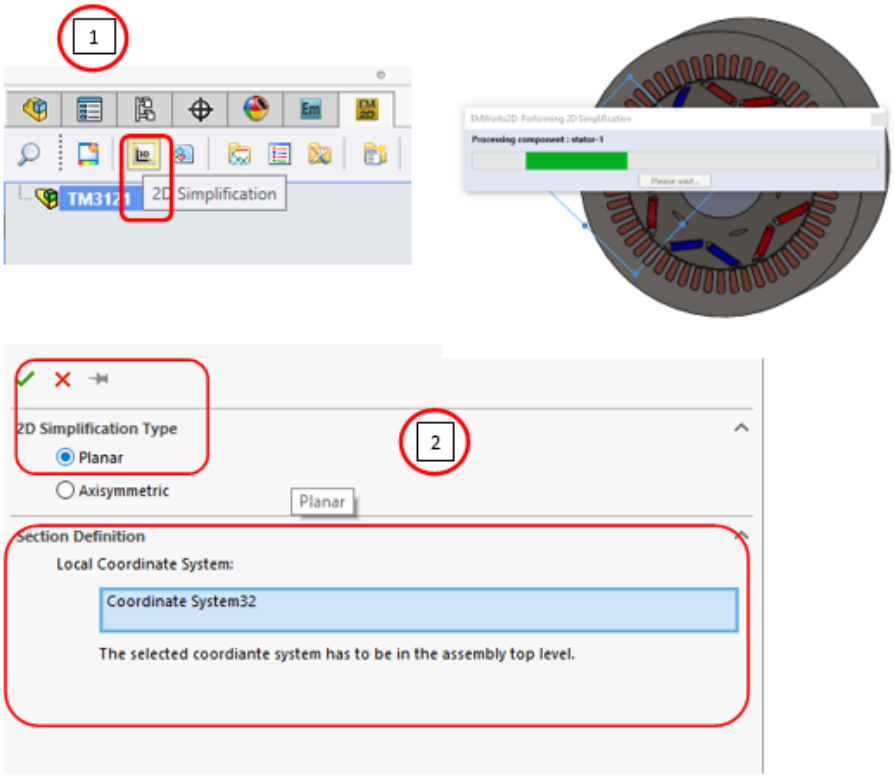


Figure 2.67: 2D planar simplification

2.5.2 Motion setup in EMworks2D

We have the rotor free to spin, we define a mate which is the angle between the rotor and stator, and we link it to an equation exactly as we did in the parametrization example above.

and then we link it to our global variable as shown

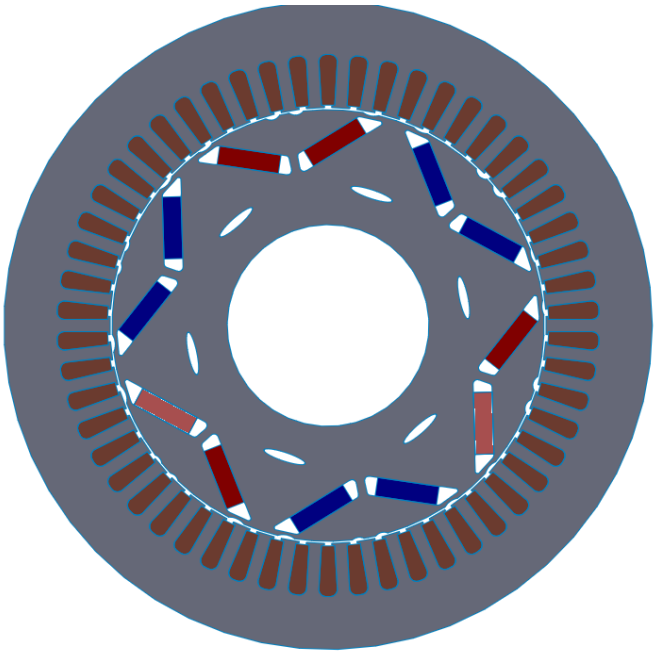


Figure 2.68: Tesla Model 3 2D model

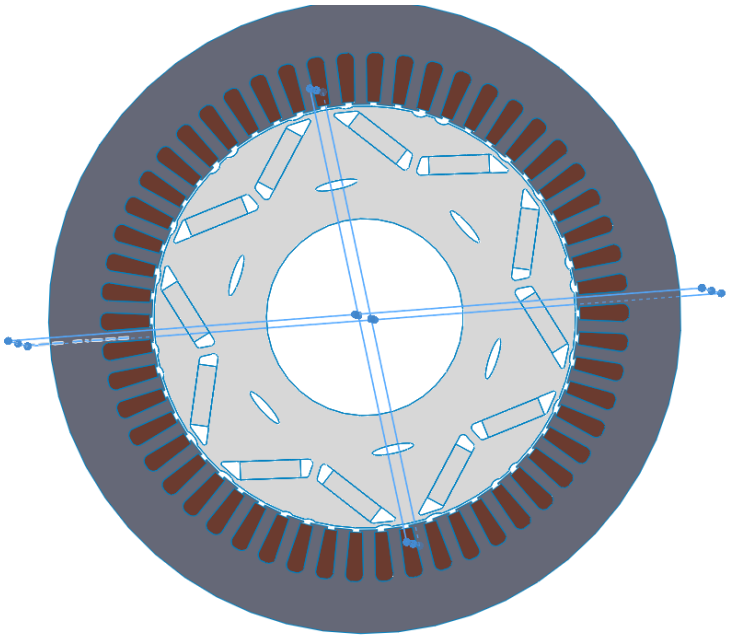


Figure 2.69: Mate definition, angle between assembly top plane and rotor assembly top plane

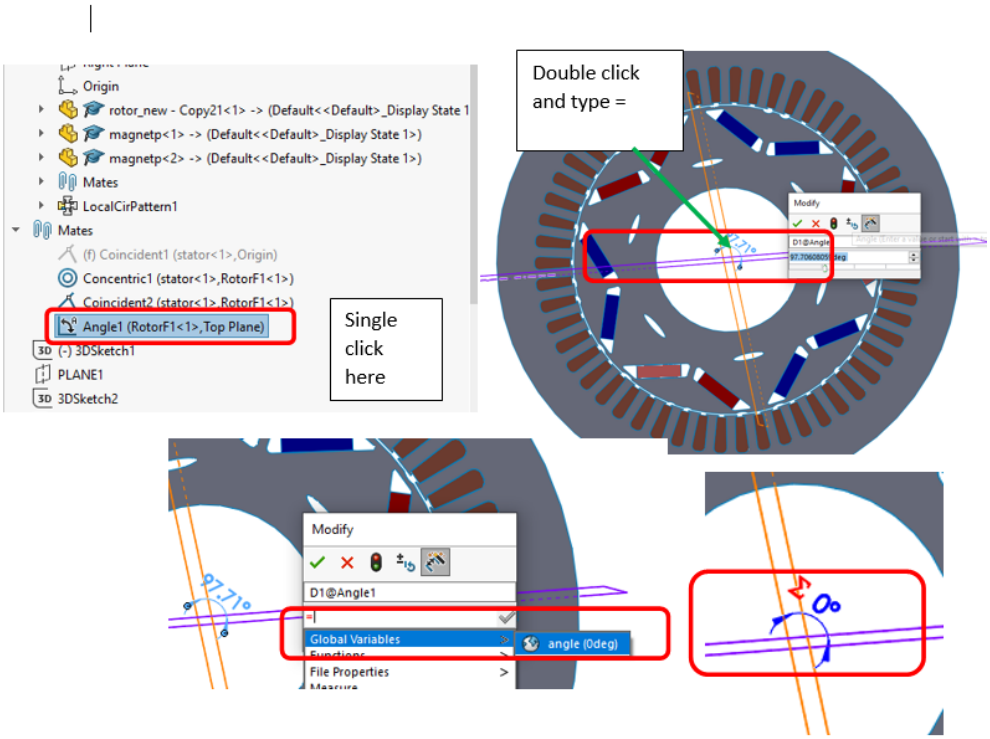


Figure 2.70: Global variable link

2.5.3 Band and air region

for Band, we insert a new component (Insert $\longrightarrow$ component $\Longrightarrow$ part, and sketch a circle that enclose the rotor and then (insert $\longrightarrow$ Feature $\longrightarrow$ surface $\longrightarrow$ planar

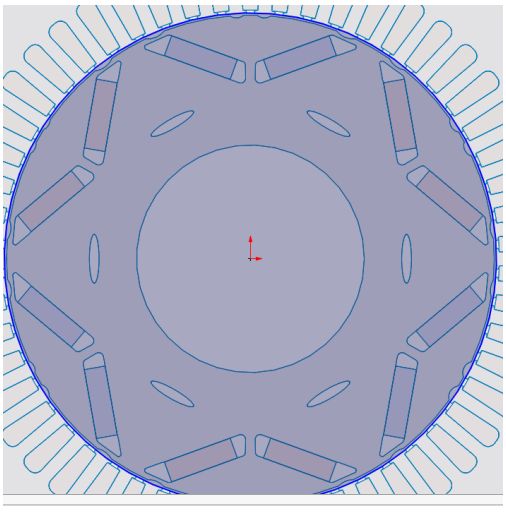


Figure 2.71: band creation

2.5.4 Transient study definition

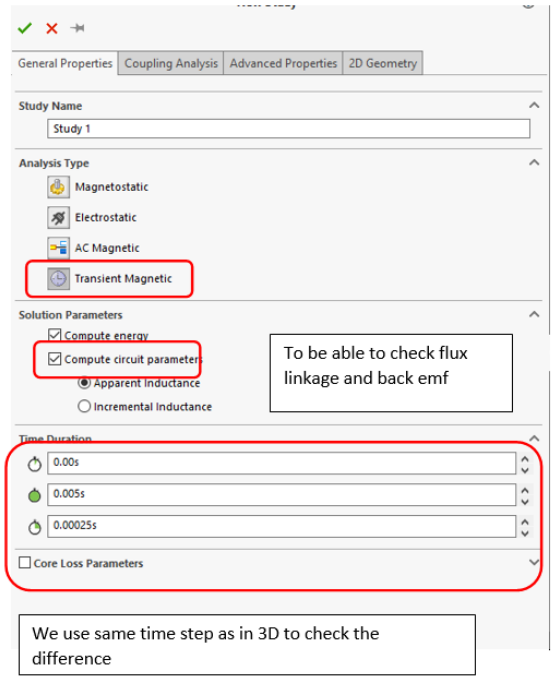


Figure 2.72: study time setup

and we need to couple to motion from coupling analysis tab

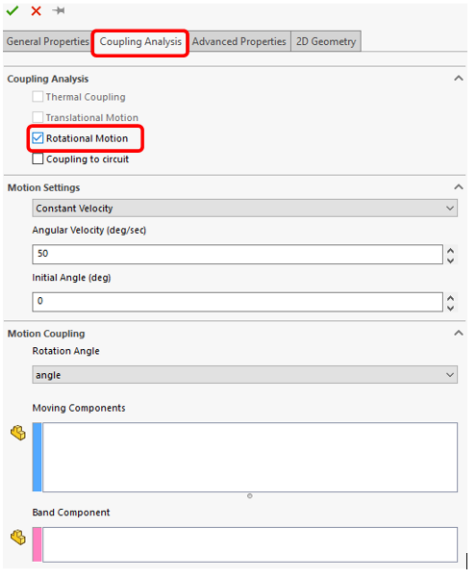


Figure 2.73: motion coupling

After setting up materials, coercivity, virtual work, and windings exactly similar to what we did in 3D, we can simulate our study, note that for speed 4000RPM is used, in EMWorks2D speed is measured with deg/s so, we used $4000 * 6 = 24000$

let's check our flux linkage and torque values



Figure 2.74: flux linkage, sinusoidal

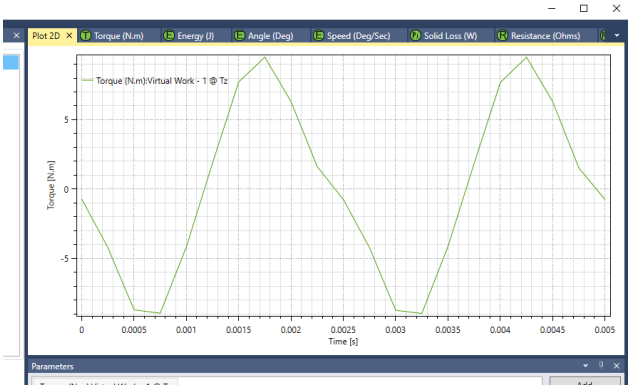


Figure 2.75: flux densuity plot

we could see that torque peak value and shape is a bit similar to 3D analysis

2.6 RZ Problem with circuit

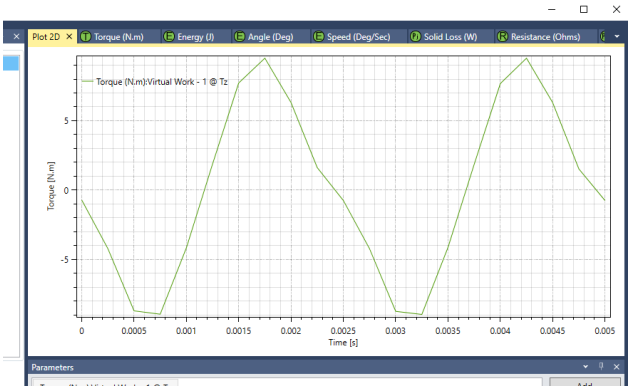


Figure 2.76: torque

2.7 Conclusion

With those basic examples, we hope that the student will gain a basic understanding about the method and about how to use EMWorks software packages